

Optical Integrated Sensing and Communication for 6G Through a Systematic Review of Architectures, Metrics, and Tradeoffs Across Optical Platforms

Fatih Dönmez, Ahmet Altuncu, and Mustafa Namdar

Abstract—Light carries data and signatures of the path and objects it encounters, allowing the same optical infrastructure to support communication and sensing. Optical integrated sensing and communication (O-ISAC) builds on this dual capability across fiber, free space optical, visible light, and light fidelity (LiFi) platforms. Photonic hardware extends the concept to millimeter wave and terahertz systems, while hybrid designs connect physical domains. This breadth motivates comparison of the design insights that extend across platforms and those tied to a particular physical setting. A systematic literature search covering 1 January 2020 through 22 June 2026 identified 227 eligible reports representing 206 unique studies. We organized this evidence by physical platform and coupling mechanism using structured narrative synthesis and a technical assessment developed for this review. The comparison framework connects each result to its task, measurement plane, operating conditions, and validation setting. Across the included corpus, bandwidth, optical power, hardware constraints, and signal design jointly shape communication reliability and sensing performance. These relationships depend on reported conditions and extend beyond a single rate and resolution frontier. Laboratory studies span diverse architectures. Broader field validation, shared benchmarks, and reusable data and code emerge as priorities for stronger comparison and reconstruction. The resulting taxonomy and framework support the design and evaluation of O-ISAC systems for sixth generation (6G) networks.

Index Terms—6G, integrated sensing and communication, metric comparability, optical communication, optical sensing.

I. INTRODUCTION

A. Background and Motivation

Communication and sensing are often needed within the same system. A mobile terminal needs a data link and knowledge of its position, while a network operator needs to carry traffic and monitor the infrastructure that supports it. Integrated sensing and communication (ISAC) brings these functions

together through shared signals, hardware, or infrastructure [1]. This interest extends to future mobile networks, with ISAC included among the usage scenarios in the International Mobile Telecommunications framework for 2030 and beyond [2].

Optical integrated sensing and communication (O-ISAC) applies this idea to optical links and photonic systems. A fiber can carry communication and sensing signals along the same route, while a visible light source can support data transmission and positioning [3], [4]. Figure 1 illustrates representative ways of combining the two functions. The role of optics differs across these systems. In fiber and optical wireless links, light propagates through the sensing medium or reaches the receiver or target. In photonics-enabled wireless systems, optical components can generate the signal while propagation to the target takes place in the millimeter-wave or terahertz band [5], [6].

Sharing a resource can also link the performance of the two functions. In one fiber experiment, raising the sensing-probe power while holding communication launch power fixed improved the sensing signal-to-noise ratio but increased the communication bit error rate [7]. Joint operation can also allow sensing to assist communication. In a visible-light prototype, position estimates guided beamforming toward the receiver and improved the communication data rate at the tested locations [4]. These examples show why the shared resource and the way it is used matter. Joint design must account for both the benefit sought and its effect on the other function.

The physical setting determines how these relationships can be used. A vibration measurement along a fiber and a position estimate in a room answer different sensing questions. Optical wireless links also depend on propagation and alignment, while fiber systems have their own transmission and sensing constraints [8], [9], [10]. A researcher selecting an O-ISAC approach therefore needs more than a list of achievable rates or sensing errors. The useful questions are which design choice produced the result, how both functions responded, and whether the conditions match the intended application. This connects the reported performance to decisions about how to build and evaluate a system.

B. Related Surveys

Existing reviews provide both platform-specific design guidance and broader views of O-ISAC. A tutorial on optical architectures and waveform allocation uses a common simulation setting to show communication error rate and ranging

This review received no specific financial or nonfinancial support. No funder or sponsor had any role in the design, conduct, analysis, interpretation, manuscript preparation, or decision to submit.

Fatih Dönmez is with the Department of Electrical and Electronics Engineering, Afyon Kocatepe University, 03200 Afyonkarahisar, Türkiye (e-mail fatih.donmez@usr.aku.edu.tr, ORCID 0000-0003-0553-4418).

Ahmet Altuncu is with the Department of Electrical and Electronics Engineering, Faculty of Technology, Afyon Kocatepe University, 03200 Afyonkarahisar, Türkiye (e-mail aaltuncu@aku.edu.tr, ORCID 0000-0002-3753-9515).

Mustafa Namdar is with the Department of Electrical and Electronics Engineering, Faculty of Engineering, Kültahya Dumlupınar University, 43100 Kültahya, Türkiye (e-mail mustafa.namdar@dpu.edu.tr, ORCID 0000-0002-3522-4608).

The corresponding author is Fatih Dönmez (e-mail fatih.donmez@usr.aku.edu.tr).

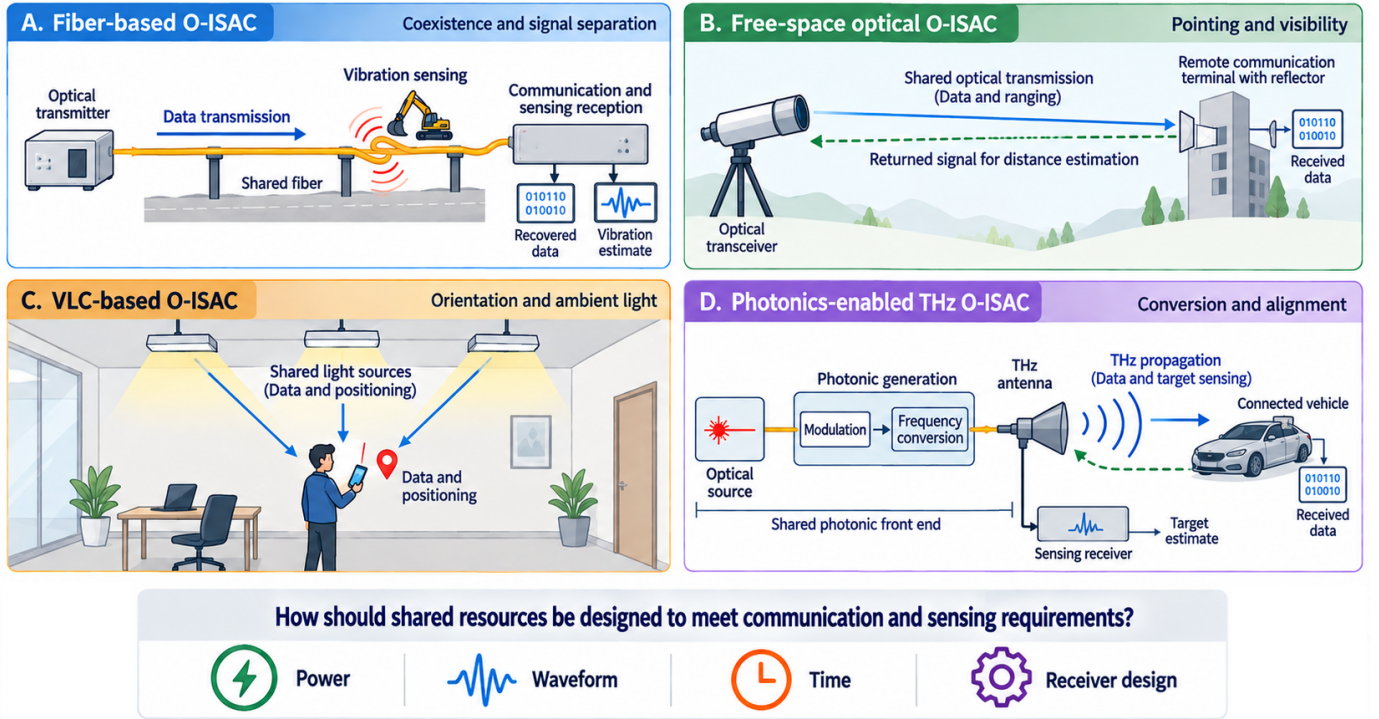


Fig. 1. Representative O-ISAC configurations in fiber, free-space optical, visible-light, and photonics-enabled terahertz systems. Each panel shows a shared resource and the communication and sensing functions it supports. The design choices below the panels can affect both functions, with their roles depending on the implementation. These examples illustrate physical configurations rather than a performance ranking or an exhaustive taxonomy.

error [11]. A review of lighting, sensing, and communication integration provides performance tables for visible-light and hybrid systems [12]. A waveform-focused review examines photonic millimeter-wave and terahertz demonstrations and compares integrated waveforms through models and experiments [13]. Other focused reviews address optical transmission and estimation, or communication and sensing in fiber networks [14], [15], [16], [17].

Broader surveys connect these technical developments to larger research questions. An evolution-oriented survey traces ISAC across spectrum, network scale, and sensing modalities, including optical and RF-optical systems [1]. A broad O-ISAC review brings together optical platforms, experimental results, waveform design, and applications [10]. These two surveys also discuss performance measures and the need for further evaluation. Together, these works explain design choices, report numerical demonstrations, and identify research directions.

Table I distinguishes the physical coverage and review approach of representative works. Its platform columns identify where the sensing interaction takes place. The last two columns address how studies are selected and how results from different studies are judged comparable. This distinction matters because a controlled waveform comparison and a comparison between independently reported experiments answer different questions. The latter also requires agreement on what was measured and under which conditions.

A further need is to explain when a reported result can guide another design. The difficulty of using common performance

measures across RF and optical systems is discussed in [1], while shared benchmarks are identified as a research need in [10]. We address this question by linking the shared design choice to communication and sensing outcomes, then examining the measurement conditions that support the link. This lets us distinguish a result demonstrated in one setting from a design lesson that can guide another. The same analysis identifies which joint measurement or experiment should come next.

C. Scope and Contributions

This survey examines 206 unique O-ISAC studies first made publicly available from 1 January 2020 through 22 June 2026. It covers guided optical systems, optical wireless links, and photonics-enabled wireless systems, including hybrid configurations. Across these settings, we examine how communication and sensing are connected through a physical resource, a signal, or a joint design objective. The review follows three questions. What is shared between the two functions? How does the shared design affect their performance? What further evaluation is needed to use the resulting insight in another system?

The survey makes three contributions that connect system selection, performance interpretation, and experimental evaluation.

- **A physical basis for selecting an O-ISAC design.** We distinguish platforms by their sensing paths and identify how hardware, signals, and allocated resources connect

TABLE I
PHYSICAL COVERAGE AND REVIEW APPROACH OF REPRESENTATIVE RELATED REVIEWS AND TUTORIAL-STYLE SYNTHESSES

Ref. / year	Main focus	Optical platforms			Review approach	
		Fiber	Optical wireless	Photonic wireless	Study selection	Cross-study comparison rules
[11] (2024)	Optical architectures, waveform design, and resource allocation	—	●	—	—	—
[12] (2024)	Lighting–sensing–communication integration and hybrid links	—	●	—	—	—
[13] (2026)	Integrated photonic waveforms and controlled performance comparisons	—	—	●	—	—
[1] (2026)	ISAC evolution across spectrum, networks, and sensing modalities	●	●	●	—	●
[10] (2026)	Optical platforms, experimental demonstrations, and applications	●	●	●	—	●
This survey	Shared-resource performance relations and evidence-based evaluation	●	●	●	●	●

●: structured treatment; ●: selected discussion or an identified research need; —: not developed in the stated sense. Fiber denotes sensing in the fiber, not fiber transport alone; photonic wireless denotes optically generated or processed signals with wireless RF propagation. Only optical-related coverage is marked for [1]. Study selection denotes an explicit search and eligibility procedure. Cross-study comparison rules address metric definitions, measurement points, conditions, baselines, and supporting evidence; they are distinct from a controlled comparison within one model or experiment.

the two functions. This explains which integration mechanisms fit a given physical setting and which constraints must be considered when adapting them (Sections II and IV).

- **A common basis for interpreting performance.** We relate bandwidth, power, waveform, and receiver choices to communication rate and reliability alongside sensing accuracy and resolution. We distinguish relationships demonstrated within a study from comparisons that require compatible metrics, measurement points, and operating conditions. This supports design lessons while keeping numerical results tied to the settings in which they were obtained (Sections II and V).
- **Evaluation priorities derived from the evidence.** We connect validation findings to joint tests under traffic, changing illumination, atmospheric variation, and optical drift. The proposed tests pair communication and sensing measurements with separate-function baselines and observable signs of progress. They give researchers concrete next steps for testing whether a design remains useful beyond its original setting (Sections VI and VIII).

Section II introduces the physical and performance concepts used in the review, and Section III describes the review methods. Sections IV through VII examine platforms, joint design, validation, and applications. Section VIII develops the research priorities, and Section IX concludes the article.

II. TECHNICAL FOUNDATIONS OF O-ISAC

O-ISAC combines data transmission and physical observation through shared optical or photonic resources. The resulting performance depends on where the signal propagates, how the receiver observes it, and which resources serve both functions. This section develops these connections through signal models, resource-sharing schemes, and numerical examples.

They provide the technical basis for reading the architectures and performance results reviewed in Sections IV and V.

A. Signal Paths and Sensing Observations

A communication receiver recovers information encoded by the transmitter. A sensing receiver estimates a property of the target or propagation medium from its effect on the received signal. The two functions can use the same received samples, different observations of one transmission, or separate signals using shared infrastructure. Figure 2 separates signal generation, physical interaction, and receiver processing. This separation locates both the sensing mechanism and the point at which each performance metric is measured.

The sensing observation depends on the physical task. Fiber vibration can change the phase or polarization of a transmitted or backscattered field. Reflective ranging uses a return whose delay depends on the target distance. Visible-light positioning can instead infer receiver location from the intensities of known illumination patterns. These paths lead to different estimators even when they share optical components [4], [7], [18], [19].

Receiver design determines which signal properties are available for estimation. For an intensity-modulated, direct-detection link, a useful linear intensity-channel model is

$$y(t) = \mathcal{R}_{\text{pd}} [h_{\theta} * x](t) + n(t), \quad x(t) \geq 0, \quad (1)$$

where $x(t)$ is transmitted optical power and $\mathcal{R}_{\text{pd}}$ is photodetector responsivity. The channel response h_{θ} depends on the parameters to be sensed, and $n(t)$ represents electrical noise. The convolution describes the intensity response within the assumed operating regime. Coherent reception instead mixes the received field with a local oscillator and provides complex observations for amplitude and phase processing [7], [11].

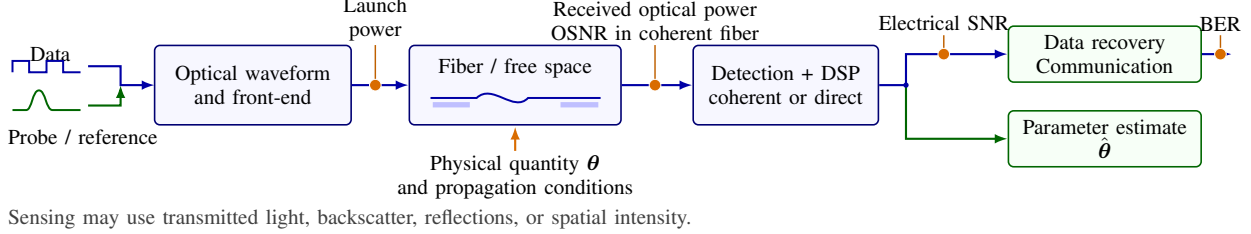
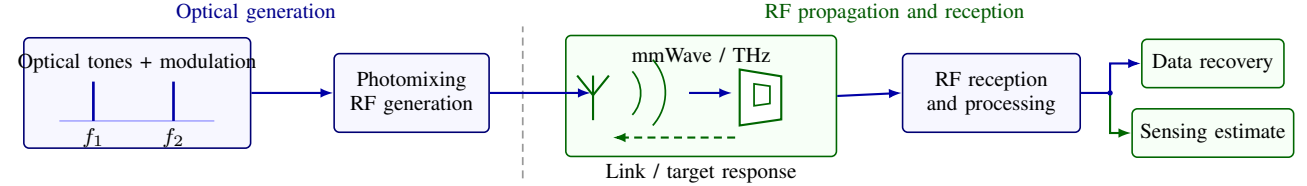
(a) Optical propagation and observation**(b) Photonics-enabled wireless propagation**

Fig. 2. Signal paths and measurement locations in O-ISAC. The upper panel distinguishes the transmitted path from sensing observations, which may use forward signals, backscatter, reflections, or spatial intensity patterns. Detection and processing depend on the architecture. The lower panel separates optical generation from radio-frequency propagation in photonics-enabled wireless systems. Optical power and OSNR are measured before photodetection, where applicable, whereas electrical SNR and recovered-data metrics belong to later stages.

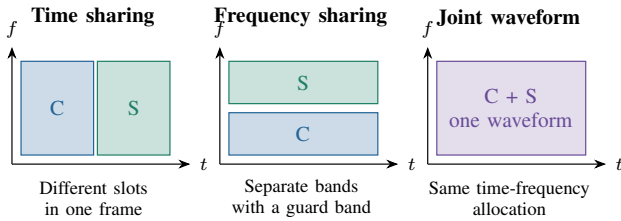


Fig. 3. Three representative time-frequency allocations. C and S denote communication and sensing use of a resource. A joint waveform reuses an allocation for both functions, whereas time and frequency sharing separate their allocations. The rectangles show resource use, not achieved performance or a ranking of integration schemes.

The role of optics changes in photonics-enabled wireless systems. Optical tones can be mixed to generate a millimeter-wave or terahertz carrier at their difference frequency. The target is then interrogated by an RF signal, although optical sources and modulators determine its waveform. For example, a 262.1–268.3 GHz transmission is generated by mixing a swept optical source with a fixed optical tone [6]. Optical-source stability and wireless propagation therefore enter different stages of the same sensing and communication chain.

B. Shared Resources and Optical Signal Constraints

Integration can reuse infrastructure, divide resources, or give one waveform two functions. Time sharing assigns communication and sensing to different slots. Frequency sharing assigns separate spectral portions, often with a guard band to limit interference. A joint waveform carries data while retaining structure that supports parameter estimation [11], [13]. Figure 3 illustrates these allocation patterns. Shared infrastructure can support any of them, while shared processing can use information from one function to improve the other.

Optical hardware constrains the waveform that can realize this sharing. In an intensity-modulated link, a bipolar electrical waveform must produce nonnegative optical power within the source’s usable range. A clipped, DC-biased model makes this constraint explicit,

$$x(t) = \min\{P_{\text{pk}}, \max[0, b + a u(t)]\}, \quad \mathbb{E}[x(t)] \leq P_{\text{av}}, \quad (2)$$

where $u(t)$ is a normalized real waveform, a sets its modulation amplitude, and b is the bias. The peak and average limits are P_{pk} and P_{av} . A larger bias can reduce lower clipping, but it consumes part of the average-power budget. The resulting balance between modulation power and distortion affects data recovery and sensing differently [11].

A DC-biased optical OFDM example uses 256 subcarriers spaced by 3.9 MHz, giving approximately 1 GHz bandwidth, and estimates range from 32-symbol frames over a 200 m FSO model [11]. Communication BER and ranging RMSE respond differently to the bias in that model. The frame structure and optical-power allocation are therefore part of the performance explanation. The nonnegative-power constraint in (2) applies to optical intensity, while complex coherent samples and the bipolar RF waveform after photonic conversion have different signal representations.

Resource sharing also permits cooperation without changing the occupied bandwidth. In a fiber experiment, the communication receiver estimates the frequency response used to pre-compensate a sensing probe. In a visible-light system, structured illumination carries spatial information while a temporal modulation carries data [4], [7]. These examples locate the coupling in receiver information and waveform structure, respectively. Their value depends on how the shared choice affects both outputs.

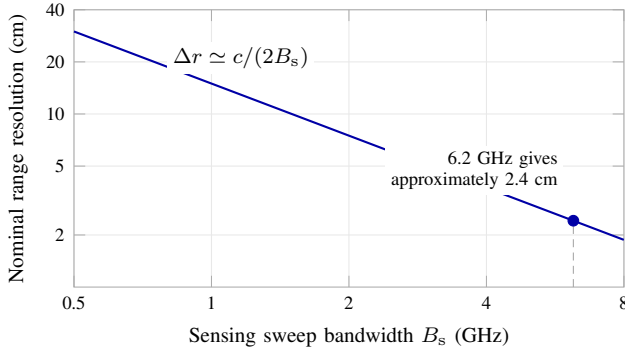


Fig. 4. Calculated nominal range resolution for an ideal two-way, bandwidth-limited measurement in air. The curve uses $\Delta r = c/(2B_s)$. The marked 6.2 GHz sweep corresponds to approximately 2.4 cm, as reported for the waveform in [6]. This is a bandwidth-based calculation, not a curve of measured estimation error.

C. Communication and Sensing Measures

Communication rate describes how information is counted over time. For a frame carrying N_{payload} useful bits over a duration T_{frame} , net rate is $R_{\text{net}} = N_{\text{payload}}/T_{\text{frame}}$. Symbol rate instead counts transmitted symbols, and its conversion to bit rate depends on modulation and overhead. Reliability adds the fraction of decoded bits in error at the stated processing stage. Thus, a data rate, a pre-correction BER, and delivered throughput describe related but different properties of a link [20], [21].

Sensing metrics distinguish target separation from parameter estimation. For two-way ranging in air, delay τ determines range r , while an ideal swept waveform gives the familiar nominal resolution

$$r = \frac{c\tau}{2}, \quad \Delta r \simeq \frac{c}{2B_s}, \quad (3)$$

where c is the propagation speed in air and B_s is the sensing sweep bandwidth. The resolution expression assumes a calibrated, bandwidth-limited delay response. Windowing, processing, and target conditions influence practical separation. Figure 4 shows this physical scale using the 6.2 GHz sweep reported in a photonic-THz experiment [6].

Estimation error answers how closely an estimated parameter matches its reference value. For N range estimates, a commonly reported measure is

$$\text{RMSE}_r = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{r}_i - r_i)^2}. \quad (4)$$

A maximum absolute error describes a different statistic. The same 6.2 GHz experiment reports a maximum ranging error of 0.55 cm for static targets at 20–45 cm [6]. This smaller error is consistent with estimating one target more precisely than the nominal separation scale for two targets. Neither number substitutes for a measured two-target resolution test.

The sensing task also determines the meaning of a spatial dimension. Fiber spatial resolution describes localization along the cable, while gauge length specifies the separation over which differential phase is evaluated. A reported 1 m spatial resolution and a 10 m gauge length therefore describe different

aspects of one fiber measurement [7]. Visible-light localization likewise separates lateral and depth performance. A demonstration resolves 1 mm lateral displacements but approximately 4 cm depth displacements in its respective tests [4].

Signal quality completes these definitions. OSNR compares optical signal and noise power in a stated reference bandwidth before photodetection. Electrical SNR depends on the detector, noise model, and electrical processing bandwidth. Launch power and received power similarly refer to different locations in Fig. 2. These distinctions explain why equal numbers in decibels need not represent equal receiver conditions [18], [21].

D. Joint Operating Points and Performance Interpretation

A joint operating point connects a design setting to communication and sensing outcomes under an identified configuration. Table II gives source-native numerical examples that make the preceding distinctions concrete. The communication and sensing columns retain their stated test settings. They show what each system demonstrates without treating different characterization tests as one simultaneous measurement.

The clearest performance link is a controlled change in a shared factor. In the fiber example, increasing sensing-probe power at fixed communication launch power raises sensing SNR while increasing the BER of both data subcarriers [7]. This is a within-study power tradeoff. The 2.4 dB sensing improvement in the table comes from a different comparison, namely probe pre-compensation using the communication receiver's response estimate. Keeping these two comparisons distinct reveals both resource competition and a benefit of cooperation.

Joint design can express such a choice as a constrained problem. For example, a sensing-oriented design may seek

$$\begin{aligned} & \min_{\mathbf{u} \in \mathcal{U}} \text{RMSE}_s(\mathbf{u}) \\ & \text{subject to } R_{\text{net}}(\mathbf{u}) \geq R_0, \\ & \text{BER}(\mathbf{u}) \leq \epsilon_b, \end{aligned} \quad (5)$$

where RMSE_s is the error in the selected sensing quantity. The vector $\mathbf{u}$ contains the shared design variables, and $\mathcal{U}$ imposes power, bandwidth, and hardware constraints. The service requirements R_0 and ϵ_b fix the communication performance that must be retained. A communication-oriented formulation reverses the objective and sensing constraint. A Pareto analysis examines settings for which improving one selected objective requires worsening the other within the feasible set [11], [13].

Interpreting results across studies adds physical and measurement conditions to this design view. Numerical comparisons require independent studies with compatible tasks, metric definitions, and measurement points. Their operating conditions and baselines must support the particular comparison, and a joint claim requires both outcomes to be linked to the shared setting. When magnitudes are not directly comparable, a recurring mechanism can still explain a design choice, with each result retained in its own context. Simulation, laboratory measurements, and field operation support different levels of validation, which Section VI examines. The study-selection method follows in Section III, while Sections IV and V use

TABLE II
REPRESENTATIVE O-ISAC CONFIGURATIONS AND THE PHYSICAL MEANING OF THEIR REPORTED PERFORMANCE MEASURES

System / ref.	Signal and test configuration	Communication measure	Sensing measure and setting	Technical distinction
Fiber [7]	40 km fiber. Two 30 GBaud 16-QAM subcarriers and a 500 MHz LFM probe. A 200 Hz vibration is applied to a 1 m fiber segment.	Reported 2×120 Gbit/s during concurrent sensing. The probe-power sweep holds communication launch power at 0 dBm.	Reported 1 m nominal spatial resolution. Differential phase uses a 10 m gauge length. Probe pre-compensation improves vibration SNR by 2.4 dB.	Span, spatial resolution, and gauge length have different roles. The 2.4 dB gain belongs to the pre-compensation comparison.
Visible light [4]	225 mW white laser diode and a 1 GHz receiver. Structured illumination covers 0.6×0.6 m at 2 m distance.	Estimated data rate averages 3.35 Gbit/s across patterns, with a 3.16–3.52 Gbit/s range.	Separate tests resolve 1 mm lateral and approximately 4 cm depth displacements. Dynamic localization operates at 39 frames/s.	Pattern-averaged rate, spatial resolving ability, and tracking update rate describe different measurements.
Photonic THz [6]	262.1–268.3 GHz joint ASK/FMCW signal. The 6.2 GHz sweep lasts 400 ns per chirp. Communication is evaluated over 1 m.	15 Gbit/s ASK with measured BER 2.97×10^{-3} . The source reports approximately 36.2 GHz total occupied bandwidth.	Nominal range resolution is approximately 2.4 cm. Maximum ranging error is 0.55 cm for static targets at 20–45 cm, using a separate receiver configuration.	Sweep bandwidth differs from occupied bandwidth. Ranging error differs from resolution, and target distance differs from communication distance.

Values retain their source definitions and evaluation settings. Nominal resolution is distinguished from measured error. The examples illustrate measurement interpretation rather than a common benchmark across platforms.

these foundations to examine architectures and joint-design results.

III. REVIEW PROCESS AND EVIDENCE BASE

With these comparison conditions established, the analysis turns to the evidence base for the technical synthesis. We assembled the corpus through a systematic review reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement. Search reporting followed the PRISMA extension for searches (PRISMA-S) [22], [23]. The following subsections describe corpus construction, evidence extraction, and technical appraisal. Table III maps the analytical role of each evidence category.

A. Corpus Construction

To assemble that evidence base, we searched Scopus, IEEE Xplore, ScienceDirect, SpringerLink, Wiley Online Library, and Taylor & Francis Online. The search covered journal articles and full conference papers that had undergone peer review and were first made publicly available from 1 January 2020 through 22 June 2026. Eligible reports contained sufficient technical content in English, examined communication and sensing within an optical or photonic system, and provided assessable evidence for at least one survey domain. Reviews and other contextual sources supported field positioning. Eligible primary reports proceeded to study reconciliation for the technical denominator. Search strategies, eligibility decisions, and protocol documentation are provided in Electronic Supplements S-Search, S-Exclusions, and S-Protocol.

After deduplication, screening, and assessment of full texts, 227 reports met the eligibility criteria. Among them, 21 were companion reports for studies also represented by another eligible report. Consolidating eligible reports by study produced 206 unique studies for the technical synthesis. The totals differ by 21 because companion reports map to studies

already represented in the eligible set. This mapping preserves provenance for every report while counting each underlying study once. Electronic Supplement ST-01 cites all 206 studies and traces their 227 reports.

Figure 5 reconciles the selection flow while keeping records, reports, and studies as distinct counting units.

The counting unit changes from records to reports at retrieval and from reports to studies at synthesis. Contextual records support field positioning, while the study set defines the technical denominator.

B. Evidence Extraction and Use

With the study denominator established, we extracted evidence at three linked levels covering the study, report, and individual claim. Each result retained its definition, unit, measurement plane, operating conditions, validation setting, and exact source location. Records remained separate when conditions differed. Each field contained a reported value or a defined availability state. Quantitative extraction used values stated by the source, while graphical findings remained in their reported form. Electronic Supplement S-Data Dictionary provides the complete coding schema.

Claims entered the analysis according to their role. Quantitative and qualitative claims supported the primary synthesis within their recorded contexts. Contextual claims and claims with source conflicts were recorded separately. This structure preserved supported evidence from all 206 studies, while comparisons across studies followed the alignment rules in Section II-D.

C. Technical Appraisal and Synthesis

To clarify how each study supports the synthesis, we assessed the technical evidence and reporting of all 206 studies with the Technical Quality Assessment Framework (TQAF) developed for this review. The framework applies deterministic

TABLE III

USE OF EXTRACTED EVIDENCE IN THE SURVEY. ROW COUNTS REPRESENT CODING RECORDS, AND STUDY COUNTS MAY OVERLAP ACROSS CATEGORIES.

Evidence use	Rows	Studies	Role in the survey
Quantitative evidence	4,997	206	Numerical or directional synthesis under recorded conditions.
Qualitative evidence	3,206	206	Mechanisms, trends, and limitations.
Claims used for context	31	21	Contextual interpretation reported separately from the primary synthesis.
Claims with source conflicts	72	31	Recorded separately under predefined decision rules, while supported evidence from the same studies remained in the primary synthesis.
Primary synthesis	8,203	206	Quantitative and qualitative evidence combined.

Note. The complete evidence register contains 8,306 rows. Electronic Supplement S-Evidence reports the 8,203 primary synthesis rows and their source locations. Electronic Supplement S-Protocol documents the decision rules.

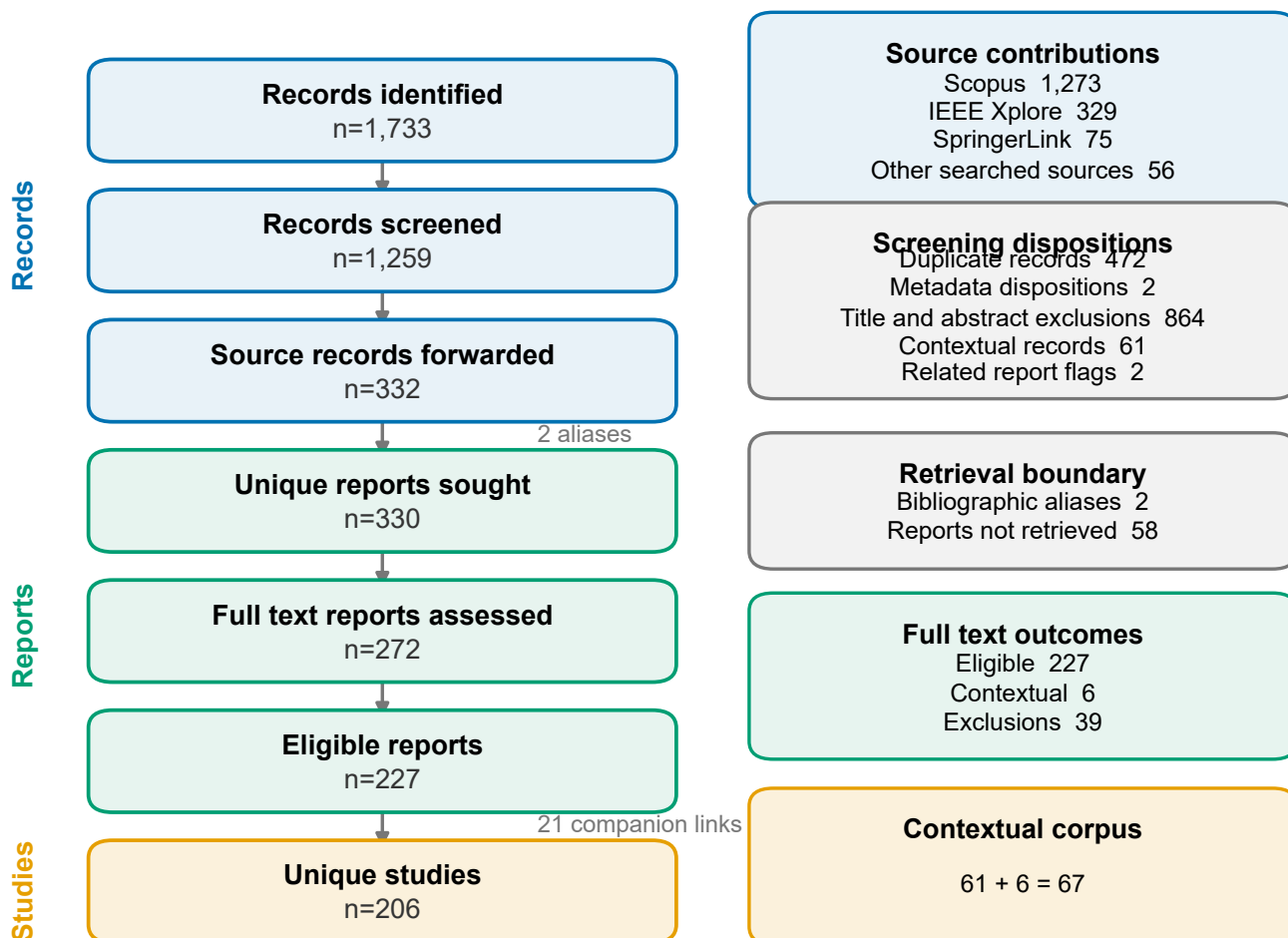


Fig. 5. Selection of the O-ISAC evidence base. Six sources yielded 1,733 records and 1,259 entered title and abstract screening. The retrieval route contained 330 unique reports after two bibliographic aliases were consolidated. Assessment of full texts covered 272 reports and produced 227 eligible reports. Companion mapping linked these reports to 206 unique studies. The contextual corpus contains 67 records that support field positioning.

rules across eight dimensions. A separate overall contribution category summarizes the support each study provides to this synthesis. The profile indicates how evidence can support interpretation in its source setting, comparison through a shared benchmark, and comparison across platforms. Electronic Supplements S-Protocol and S-Appraisal provide the framework's methodological scope, scoring rules, and records for each study.

Figure 6 places the eight appraisal dimensions beside the

separate overall evidence contribution category. The profile distinguishes support for source interpretation from support for transfer and reuse.

Reporting completeness and metric clarity support interpretation within the source setting. Validation, reproducibility, benchmark readiness, and comparison admissibility introduce further conditions for transfer across studies. These distinct roles explain the profile and guide the structured synthesis.

The structured narrative synthesis retained each result's

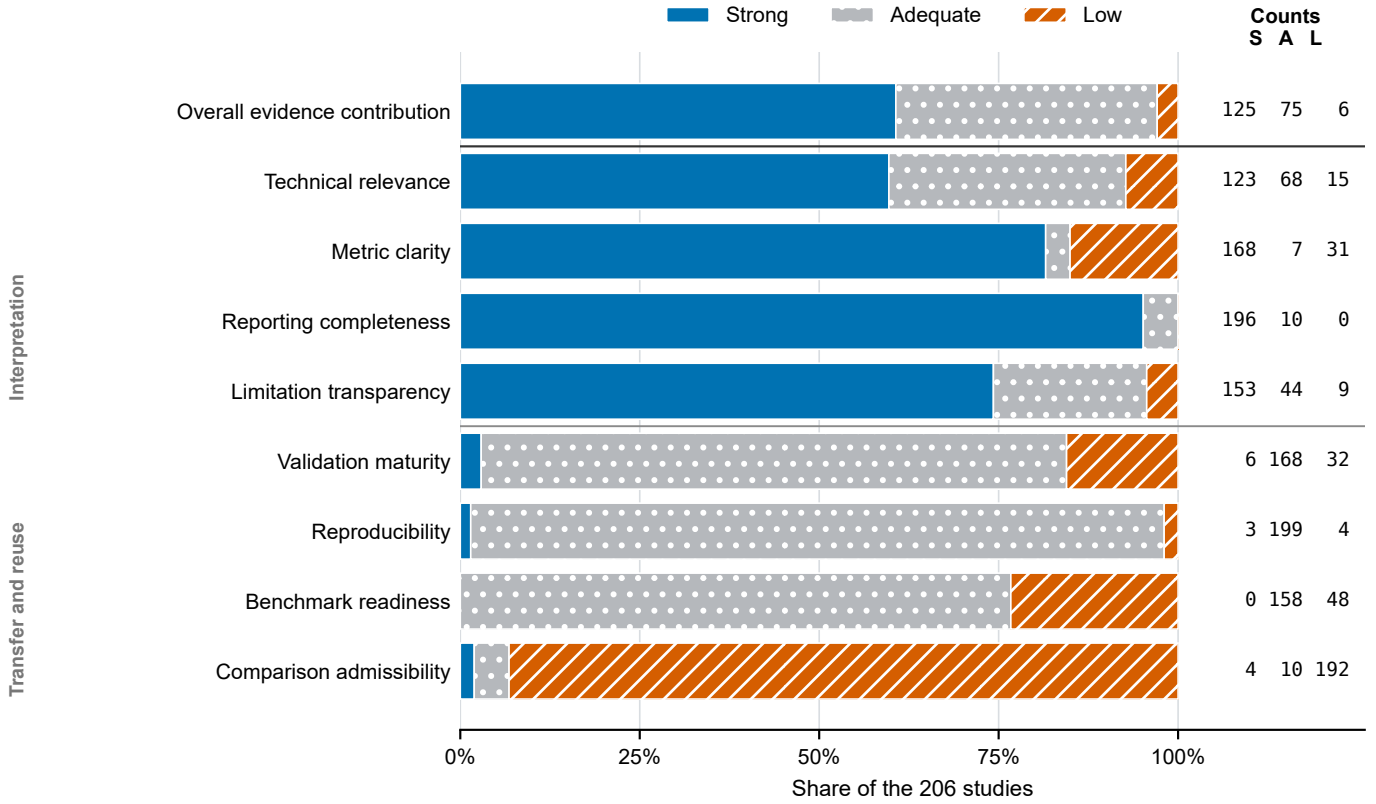


Fig. 6. Technical appraisal profile of the 206 included studies. Each row assigns all studies to strong, adequate, or low evidentiary support under the TQAF rules developed for this review. The interpretation group covers technical relevance, metric clarity, reporting completeness, and limitation transparency. The transfer and reuse group covers validation maturity, reproducibility, benchmark readiness, and comparison admissibility. Overall evidence contribution remains a separate summary category.

native task and unit together with its operating conditions and validation setting. This structure preserved the meaning of heterogeneous results while allowing recurring mechanisms and design relationships to be compared across the corpus. The evidence was organized into 115 bodies across seven survey domains, connecting synthesis conclusions to their supporting studies. Electronic Supplement S-Bodies provides their criteria, ratings, and memberships. With this evidence structure in place, the technical synthesis begins with the physical role of optics and the mechanism that couples communication and sensing.

IV. OPTICAL PLATFORMS AND INTEGRATION ARCHITECTURES

In O-ISAC, the physical role of optics determines both where information propagates and where the environment is observed. Optics may carry the user link, illuminate a target, distribute a wireless carrier, or provide a coherent reference, and each role exposes a different set of impairments. The integration architecture then reveals which component, signal, or resource is shared by communication and sensing. We therefore examine the evidence along the axes of physical platform and coupling mechanism.

Table IV reports the mutually exclusive family counts and corpus shares. Photonic terahertz and fiber together account for 125 studies, representing 60.7% of the corpus.

TABLE IV
CANONICAL PHYSICAL ORGANIZATION OF THE CORPUS OF 206 STUDIES

Platform family	Studies, <i>n</i> (%)
Photonic terahertz	69 (33.5%)
Fiber	56 (27.2%)
Visible light communication and LiFi	38 (18.4%)
Free space optical	31 (15.0%)
Hybrid optical	9 (4.4%)
Other optical	3 (1.5%)
Total	206 (100.0%)

Note. Each study is counted once under its dominant platform family. Media and integration mechanisms can overlap within the represented systems. The counts summarize how the reviewed evidence is distributed across the six platform families.

The family profiles below provide the corresponding physical paths, observables, coupling mechanisms, active constraints, transfer conditions, and evidence anchors. Each study belongs to one platform family, while its coupling mechanisms may overlap. Transmitters supplied through fiber enter the terahertz family when photonic generation and conversion create the wireless carrier directed at the target. Fiber remains the organizing platform when it transports signals to downstream radio frequency identification (RFID), radio, or radar sensing. Hybrid membership begins when interaction between physical segments shapes the result. This organization locates the target

path and guides the synthesis that follows.

A. Physical Platform Families

1) *Photonic Terahertz Systems*: Photonic terahertz is the largest family, comprising 69 of the 206 studies (33.5%). The optical chain generates, distributes, or processes a coherent carrier before conversion to a wireless terahertz or millimeter wave target path. It yields delay, range, Doppler, angle, imaging, and channel or target response observables. Communication and sensing are coupled through hardware, references, waveforms, beams, or resources [5], [6], [20]. This upstream sharing ties both functions to the same optical conditions. Laser coherence and linewidth govern reference stability, while component bandwidth, conversion fidelity, and efficiency shape the bandwidth, phase structure, and power delivered to the antenna [24], [25]. At the radio interface, antenna aperture, gain, and alignment govern launch and collection, molecular absorption shapes propagation, and target scattering forms the returning echo in radar configurations [26], [27].

Where the functions meet varies by architecture, with some systems dividing a multicarrier frame between sensing and communication and others placing the joint decision in a phased array controlled through photonics [5], [20]. Wavelength tuning and multibeam reception move coordination to the source or the receiving front end [28], [29]. At the waveform level, one multichannel design combines linear frequency modulation (LFM) with communication symbols across coherent carriers [24]. Other designs add source agility through frequency hopping or organize the signal over delay and Doppler through orthogonal multiplexing [27], [30]. Information can also close the loop, as burst SNR estimates guide both probabilistic shaping and ranging, while a simulated radar waveform carries communication data [26], [31]. Together, these designs show how the joint decision can move from hardware to waveform and to the information used by both tasks.

The location of that decision also determines how the measurement is formed. A returned echo may be remodulated onto light and carried back over fiber for centralized processing, where photonic dechirping expresses target delay at a lower frequency [27], [32]. Along this path, phase noise associated with laser linewidth degrades the recovered signal, while delay mismatch between optical and electrical paths shifts the ranging observable [32]. Laboratory experiments reveal waveform, conversion, and beamforming behavior under controlled conditions [5], [24], whereas a 270 m outdoor link adds wireless propagation and operation of the complete chain [20]. Comparison across photonic terahertz systems depends on aligned optical and wireless stages, aperture and beam conditions, receiver processing, and validation setting.

2) *Fiber O-ISAC*: Fiber comprises 56 studies (27.2%) and follows two physical paths. The guided link may form the sensing medium, with disturbances observed through backscatter, carrier phase, modal coupling, or an inserted sensor [33], [34], [35], [36]. Fiber may instead transport signals to a downstream RFID, radio, or radar segment, so the target interaction

occurs after optical distribution [37], [38], [39], [40], [41]. The corresponding observations include distributed events, vibration, strain, and environmental change in the guided path, together with downstream tag state, range, localization, and imaging [33], [37], [38]. The measurement boundary therefore lies either within the guided path or across the complete optical and wireless chain.

Coupling recurs through shared hardware, wavelengths, waveforms, links, or resource decisions. Integration can reuse the communication signal itself [34], [35], [42], [43], [44] or place distinct data and sensing signals on shared infrastructure [3], [33], [36], [45], [46]. The first form ties sensing to the symbols, carrier, phase, or supported modes, whereas the second ties both functions to the same fiber and receiver conditions. Guided sensing is constrained by dynamic range, leakage, reflection, nonlinearity, reference stability, topology, and traffic, whereas transported sensing additionally depends on conversion, timing, remote calibration, and wireless geometry. Evidence comes from laboratory links, live access networks, and a deployed submarine route. For fiber evidence, comparison depends on the interrogation or transport role, topology, launch and receiver conditions, conversion and clocking, target path, traffic, and validation setting [33], [37], [38], [41], [42], [43], [46].

3) *Visible Light Communication and LiFi Systems*: Visible light communication and LiFi comprise 38 studies (18.4%) and place communication and sensing within the same illuminated scene. The grouping preserves their distinct meanings at the link and networking levels. Direct or reflected room and device paths through emitters, cameras, and photodiodes support position, pose, distance, imaging, gesture, recognition, and physiological sensing [4], [47], [48]. Cameras and photodiodes expose different balances between spatial detail and reception speed, while nonnegativity, dimming, flicker, source linearity, ambient light, field of view, and exposure constrain what can be communicated and observed [47], [49], [50], [51]. Coupling recurs through hardware, waveform, geometry, resource decisions, and the application loop. It may arise through a shared observation [48], [52], through geometry and resource decisions [53], [54], [55], [56], [57], or through integrated emitters and detectors [58], [59], [60].

Illumination is the common comparison boundary across these designs. Geometry and multipath shape localization, reflected observations support gesture and recognition, and learned models extend the same optical field to health sensing tasks [47], [51], [55], [56], [61], [62]. Analytical and simulated studies isolate geometry and resource choices, whereas prototypes and field experiments add lighting, receiver, and control conditions [47], [52], [53], [55], [57], [59]. Visible light comparisons depend on matched task definitions, layout, orientation, illumination, detector, processing, and validation conditions.

4) *Free Space Optical Systems*: Free space optical systems comprise 31 studies (15.0%) and use a directional link, optionally combined with target illumination and return, over direct or reflected aperture and beam paths. These paths support ranging, localization, imaging, tracking, and atmospheric or channel observation [63], [64], [65]. Geometry is therefore

part of both the link and the measurement. Coupling recurs through the link, waveform, aperture or beam, and spatial or temporal resources, while pointing, divergence, aperture, reflectivity, loss, turbulence, and motion constrain the resulting evidence. Direct, reflected, scanned, and steered paths illustrate these relations [63], [66], [67], [68]. Retroreflection and multicarrier reception connect returned power to range and position [64], [65]. Structured waveforms and steering apertures place coupling in the signal or beam [69], [70], [71], [72], while atmospheric estimates can guide beam shaping or link selection [9], [73].

Across these forms, beam geometry determines both spatial selectivity and the power collected at the receiver. Range, angle, and image features therefore remain tied to calibration, aperture, pointing, and motion [63], [64], [65], [66], [67]. Controlled links reveal waveform and estimator behavior, whereas operational ranging adds atmospheric variation and continuous field observations [9], [65]. Comparison across free space optical systems depends on matched geometry, calibration, target properties, weather, motion, and validation conditions. Hybrid systems extend that comparison across interfaces between physical segments.

5) *Hybrid Optical Systems*: Hybrid optical systems comprise nine studies (4.4%) in which consequential interaction occurs among optical, fiber, free space, or radio segments and the perturbation and observation may straddle a conversion boundary. Their observables include range, vibration, and target or network state within individual segments, while coupling arises through a central reference and through links, resources, control, or applications shared across segments. Conversion loss, segment power, synchronization, calibration, and timing are therefore active constraints [74], [75], [76]. Hybrid comparisons trace every segment, conversion, clock, and measurement plane together with the validation of both functions.

Unlike fiber used only for downstream transport, hybrid membership requires conversion or feedback across segments to shape the result. This occurs when interaction between fiber and wireless segments is consequential [74], [75], when a shared photonic source spans fiber, free space, and millimeter wave stages [77], or when sensing in one segment guides service in another [76], [78]. Other designs establish a common signal origin through shared time modes or a photonic waveform with frequency agility [79], [80].

6) *Other Optical Systems*: The three other optical studies (1.5%) use distinct mechanisms based on optical control or coherent carriers. These comprise a microwave field generated by a microcomb, a solar blind ultraviolet path, and a quantum optical path. Their observables are, respectively, a field pattern, particulate or smoke response, and phase [81], [82], [83]. Coupling is specific to specialized hardware, reference, and application choices, and each case is bounded by its own propagation, target, receiver, and measurement models. Each numerical result is therefore interpreted within the native mechanism and measurement model of its system. Together, the hybrid and other optical cases complete the physical map before the analysis turns to coupling mechanisms shared across families.

B. Integration Architectures and Resource Coupling

Across these physical families, integration architecture locates where communication and sensing meet along the target path. The coupling may enter through shared hardware, a common carrier or waveform, coordinated resources, a propagation link, or a service decision. Several mechanisms can coexist in one system. Each label identifies one coupling location while preserving the rest of the chain for separate interpretation.

Figure 7 locates seven recurring coupling mechanisms along a generic signal and service path. A study may occupy several locations, so the badges report overlapping coverage across the corpus.

Resource, hardware, and waveform sharing can form one connected design chain. Application coupling acts at the service level and carries a different engineering meaning. The following subsections trace these locations from physical reuse to coordinated resources and design.

1) *Shared Hardware, Carrier, and Link*: Hardware, carrier, and link sharing describe different physical relations. Reusing a component makes its bandwidth, coherence, dynamic range, or alignment conditions relevant to both functions. A common optical carrier adds a phase or wavelength relation, while a common link adds shared propagation and calibration conditions. Together, these relations show which design choice is common and which constraint follows it.

The physical path gives reuse its engineering meaning. In fiber, traffic and sensing share the guided structure, with a resonance shift, phase, or backscatter conveying the sensed change [3], [84]. A retroreflective free space path supports information delivery together with ranging or localization [63]. Arrays controlled through photonics instead share optical generation and beam control while the target path remains wireless [5]. Comparison therefore follows the component role, carrier relation, propagation path, and calibration.

2) *Shared Waveforms*: Waveform sharing begins when one signal structure contributes to both decoding and sensing. The shared element may be the complete waveform or a selected pilot, chirp, code, polarization state, or spatial pattern. This common design links payload efficiency, sensing observability, and receiver processing. Structural sharing records that relationship. Concurrent performance adds the requirement that both outcomes be observed at a matched operating point, a distinction carried into Section V.

The evidence illustrates several forms. Pilots inserted for channel estimation can also provide a sensing reference [85]. One photonic terahertz prototype generates a waveform whose frequency steps provide the radar dimension and whose amplitude carries data. Its experimental evidence is strongest for waveform generation and communication response [86]. A visible light design combines temporal data with spatially structured illumination and uses localization to adapt beam-forming [4]. The shared structure is similar, while its physical and receiver implementation remains specific to the platform.

3) *Resource Allocation and Joint Design*: Resource coupling coordinates temporal, spectral, power, spatial, or processing degrees of freedom. An allocation can change payload opportunity, link margin, sensing observability, geometry, or latency. Joint design connects the two functions more directly

Coupling locations may overlap within one study

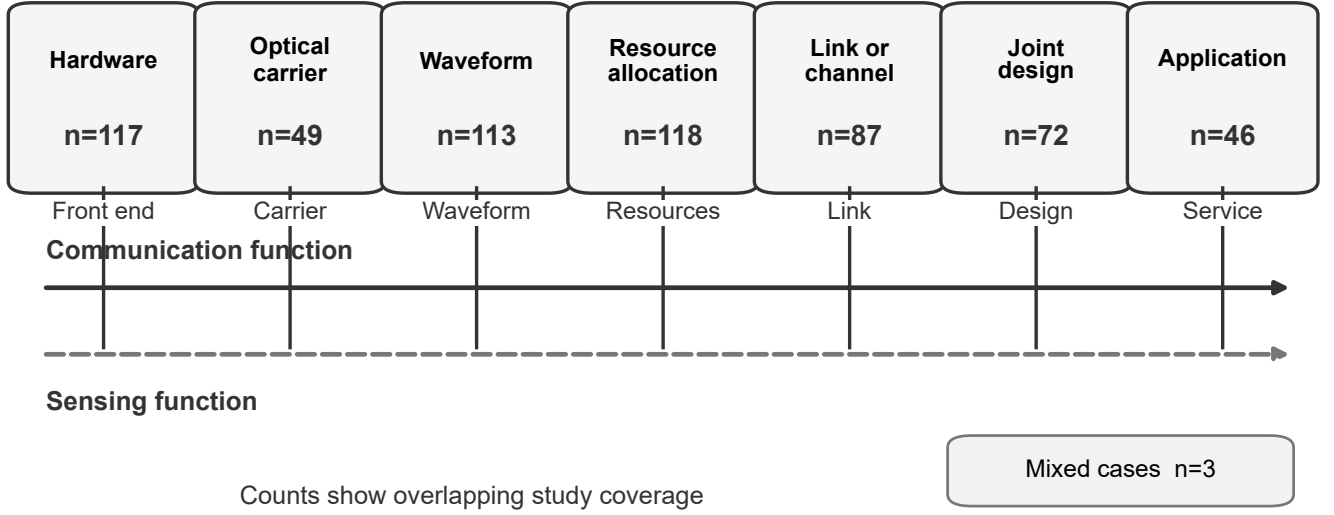


Fig. 7. Locations of communication and sensing coupling in the corpus of 206 studies. Equal cards identify hardware, optical carrier, waveform, resource, link, joint design, and application coupling along the system path. The badges report overlapping study coverage, with three mixed boundary cases shown separately. Card position identifies the typical coupling location, while each card retains equal visual weight.

by choosing a variable, objective, or constraint in view of both outcomes. The resource, allocation rule, and operating conditions therefore define the resulting evidence.

A visible light model for vehicle to everything communication makes exposure time a shared decision variable. Its coexistence result is analytical and numerical, while laboratory communication and sensing gains were obtained in separate configurations [49]. In one reported terahertz configuration, a rapidly swept photonic source and common waveform plan support simultaneous data transmission and ranging [6]. These cases distinguish a modeled optimum from a measured concurrent configuration. Section V therefore reads each relationship within its operating region and evidence form.

4) *Coupling at the Application Level and Boundary Cases:* Application coupling arises when sensing informs beam or link selection, trajectory planning, blockage response, or network control. The shared element is then the system action, even when the signal paths remain separate. A near infrared wearable combines communication and physiological sensing on a common material platform while retaining separate detector roles [60]. A hybrid last mile design places free space optical delivery and fiber Bragg grating monitoring within one system model [76]. Their coordination lies in hardware and service context.

Three studies cross temporal or physical segment boundaries. One performs localization and imaging before data delivery on a shared optical source and path with separate receivers [66]. Two use sensed free space optical channel state to adapt trajectory and resource allocation or cell placement and user association across optical backhaul and radio access [87], [88]. The mixed label preserves this path and keeps each

result tied to the segment, function, and operating stage that produces it.

C. Interpretation Across Platforms

Read together, the platform and architecture views form one mechanism map. The platform establishes the physical transfer conditions, while the architecture identifies the shared choice. In fiber, waveform sharing inherits guided propagation and reflections, together with launch and receiver conditions [42], [43]. In free space optical systems, the same mechanism is shaped by aperture, geometry, and alignment, whereas terahertz systems assisted by photonics also introduce conversion from optics to radio [5], [63]. Application coupling can instead coordinate separate optical and radio paths through a common service decision [87].

Across these mechanisms, sharing concentrates design around a common component, reference, or resource. That choice can couple the functions through power balance and interference, while synchronization and receiver dynamic range constrain their joint operation [3], [75]. Section V therefore examines which communication and sensing outcomes change together at a matched operating point and which conditions explain that relationship.

V. PERFORMANCE METRICS AND JOINT DESIGN TRADEOFFS

After locating a shared choice, the analysis turns to the reported communication and sensing outcomes. A reported value gains engineering meaning from its task, its observation point in the signal chain, and its operating and validation

conditions. The comparison profile connects these elements to the performance evidence.

The primary synthesis contains 4,779 metric records. Each record is an extraction unit that retains the reported quantity together with the definition and source conditions available for it. The denominator therefore represents reported quantities in their original contexts. Data rate, bit error rate (BER), and range resolution have the widest study coverage, appearing in 85, 60, and 50 studies, respectively. Since one study may report several families, these totals provide overlapping views of the evidence.

Beyond these familiar measures, the evidence extends from link and waveform behavior to outcomes observed at the device, estimator, application, and implementation levels. This breadth makes semantic alignment the organizing principle for the analysis that follows.

A. Applying the Comparison Profile to Individual Metrics

Applying the comparison profile defined in Section II, each metric row was coded by source definition, task, measurement plane, operating point, validation method, and value provenance. Communication and sensing results form a joint operating point when the source links them to the same condition set. Table V summarizes domain coverage and the analytical role assigned to each record.

TABLE V
METRIC EVIDENCE PROFILE FOR 4,779 PRIMARY RECORDS

<i>Metric domains</i>		
Domain	Records, <i>n</i> (%)	Studies, <i>n</i> (%)
Sensing	1,816 (38.0%)	203 (98.5%)
Communication	1,328 (27.8%)	194 (94.2%)
Joint	870 (18.2%)	158 (76.7%)
Implementation	476 (10.0%)	64 (31.1%)
Smaller normalized domains	289 (6.0%)	33 (16.0%)
<i>Analytical roles</i>		
Role	Records, <i>n</i> (%)	Use in the survey
Bounded relation	118 (2.5%)	Conditional comparison across studies
Native context	4,661 (97.5%)	Mechanism, feasibility, and qualitative synthesis

Note. Domain rows and analytical roles each partition the 4,779 primary records, while study counts overlap. The complete metric register contains 4,861 rows. The additional rows comprise 31 contextual records and 51 records documenting source conflicts. Comparison follows the reported definition, task, measurement plane, conditions, and validation. Electronic Supplement S-Evidence provides the complete records and provenance, including quantities grouped under the broader other category.

The domain distribution shows where metric evidence is concentrated, while the analytical role determines how each record enters the survey. The following subsections explain how task, measurement plane, and operating conditions shape communication, sensing, joint, and implementation metrics.

B. Metrics Across Functions

1) *Communication Metrics:* A communication rate gains meaning from what the source counts. Symbol rate tracks signaling pace, while data rate reflects the payload and overhead

included in that count. Throughput describes delivered service, capacity provides a model bound, and spectral efficiency relates useful data to its stated denominator. In a terahertz prototype enabled by photonics, net rates follow communication subcarrier allocation, payload occupancy, and coding overhead [20]. An embedded pilot orthogonal frequency division multiplexing (OFDM) simulation defines spectral efficiency as the data subcarrier fraction and reports cyclic prefix overhead separately [85].

Reliability shifts attention to the event and its location in the signal chain. BER measured before and after forward error correction describes two stages of the link, while packet loss and outage describe broader delivery boundaries. A fiber experiment over a long span reports receiver OSNR and BER before correction during concurrent distributed sensing [21]. A robotics study reports packet loss over distance and view angle, display delay over queue size, and formation control under dynamic motion as separate results [47]. The reported reliability value is therefore tied to its event, observation point, time scale, and system boundary.

The observation point also determines how signal quality and power are interpreted. OSNR relates signal power to noise in a stated optical reference bandwidth, while electrical SNR follows photodetection and digital SNR follows equalization. Launch and received power likewise refer to the transmitter and receiver ends of the link. A fiber polarization sensing experiment traces recovered vibration SNR across an OSNR sweep, connecting the sensing response to a stated optical condition [18].

Distance and latency complete the profile by locating the result in space and time. Fiber span and wireless separation describe link geometry, while target range and coverage radius describe the observed or served region. Latency may include propagation and buffering, signal acquisition and processing, or the full application loop. Stated endpoints, included stages, and operating conditions define the scope of each value. The complete profile connects each communication metric to the system behavior it represents.

2) *Sensing Metrics:* Sensing metrics first take meaning from the task and quantity they represent. Range and position estimates, motion measurements, images, and class decisions refer to distinct observations and references. Resolution describes separability along a stated dimension, while estimation error relates an estimate to a reference. Repeatability captures variation across trials, and a model bound supplies a conditional benchmark. These distinctions establish the comparison basis.

Representative studies show how this contract works. A fiber reflectometry system reports breakpoint localization error and range resolution separately for laboratory and simulated conditions [89]. A visible light localization experiment distinguishes lateral estimation error from the smallest demonstrated displacement and summarizes signed displacement with means and standard deviations across repeated trials [4]. An optical phased array simulation relates imaging root mean square error to electrical SNR under a defined scene, waveform, and beamforming constraint [90]. Each result retains its summary, geometry, and validation setting.

Detection and classification add dataset and decision context. Their metrics take meaning from class balance, decision thresholds, background conditions, and the relation between training and test data. CORE-Lens reports recognition accuracy and optical camera communication BER across stated class, data rate, and background settings, with runtime evaluated separately [91]. VehicleTalk evaluates link performance, distance estimation error, and vehicle status recognition under separate prototype conditions [48]. Their application meaning follows from the attached classes, geometries, and test protocols.

Physical quantity sensing follows the same logic by linking each measurand to a reference and calibration. Spatial support and acquisition time locate the observation, while the aggregation rule shapes its summary. The resulting profile connects the task, reference, geometry, processing, and validation setting to the reported sensing outcome. Together, these elements define the conditions for comparison.

3) *Joint and Implementation Metrics*: Joint metrics connect communication and sensing through a shared operating point or resource rule. A matched tuple uses the same conditions for both functions. A shared resource can also link values that retain their own reported conditions. In an embedded pilot OFDM simulation, BER and mean absolute pilot error are reported at the same SNR with eight pilots, while spectral efficiency follows the corresponding data subcarrier fraction [85]. The pilot error measures deviation of known pilot symbols in the simulated channel. Together, the three values show how pilot allocation connects communication reliability, resource efficiency, and the reported sensing measure.

Optimization creates another form of joint evidence by selecting an operating point through shared design variables. In a simulated optical phased array system, precoding matrices and photodiode orientations are selected together. The design maximizes a sensing contrast metric under a communication quality threshold and emitter power constraints [90]. Geometry, constraint thresholds, and metric definitions define the scope of the solution.

Implementation metrics place these operating points within the platform and application chain. Receiver bandwidth, converter rate, memory, acquisition time, and latency show where processing burdens arise. An outdoor photonics assisted terahertz prototype evaluates receiver pipeline latency in hardware analysis and communication and ranging performance in a field experiment [20]. A smartphone camera prototype evaluates recognition and communication in the laboratory and benchmarks module latency, memory, and complete cycle timing on the device [91]. Together, these measures connect joint performance to the processing chain and locate the effects of shared design choices in system behavior.

C. How Communication and Sensing Performance Interact

Communication and sensing often share bandwidth, power, waveform components, hardware, and processing. A design change in any of these elements can move both functions to a new operating point. More pilot symbols may strengthen a sensing reference while leaving fewer resources for data.

Wider bandwidth may support both a higher data rate and finer range resolution until a receiver or platform limit becomes active. The evidence is organized by the shared choice that produces each pattern.

The synthesis covers 402 relationships from 168 studies. Figure 8 separates record type, overlapping study coverage, and condition dependence across 11 recurring mechanism families. Each relationship remains tied to its waveform, geometry, target, observation point, and validation setting.

Resource and power mechanisms carry much of the coded evidence, while condition dependence remains common across the families. This pattern directs the discussion toward the shared mechanism and the operating condition that shapes each reported relation.

1) *Bandwidth, Spectrum, and Resource Allocation*: Bandwidth links the two functions when both draw on the same signal space. That space may be divided across subcarriers, wavelengths, guard intervals, observation time, or the usable response of a device. Embedded pilots show the basic pattern. Increasing pilot support improves the reported pilot deviation proxy while changing BER and the data carrying fraction [85]. Protection intervals and chirp sweeps create related exchanges by improving sensing isolation or pulse compression within device bandwidth and subcarrier limits [45], [75]. In each case, the effect follows the resource being shared and the limit that becomes active.

Waveform structure determines how the shared signal space is used, with format and operating mode setting time and spectral sharing [6], [63], [66]. Within that structure, digital designs assign subbands, chirp parameters, power, and weighted resources [92], [93], [94], [95], [96]. Pilot strength, placement, and synchronization fields define reference support and communication margin, while receiver partitioning and ranging sequence redundancy set the usable signal region [20], [97], [98], [99], [100], [101]. A tunable laser study supplies qualitative generation context for the same design space [102]. Together, the allocation rule and receiver condition define the operating point reported in each design.

Resource sharing also appears in time. Observation duration, camera exposure, and task scheduling connect sensing quality and cadence with communication goodput or execution time [49], [91], [103]. Time division, chirp repetition, coding traffic, security control, and embedded pulse count place both functions within a shared frame [48], [52], [104], [105], [106]. Other designs reserve spectral, temporal, spatial, polarization, or phase structure to manage interference, coverage, and receiver complexity [8], [29], [40], [107], [108], [109]. The observed outcome therefore depends on the frame period, exposure response, isolation rule, and remaining occupancy.

When both functions occupy the same waveform, bandwidth affects pulse compression, estimation precision, sidelobes, and radar image quality together with communication support [110], [111], [112], [113], [114]. Stimulated Brillouin scattering, frequency partitioning, and constellation choice reveal device and spectral limits [24], [115], [116]. Doppler refinement depends on phase noise and computation [117]. A communication and sensing bandwidth ratio gains meaning from total bandwidth, target range, and receiver architecture

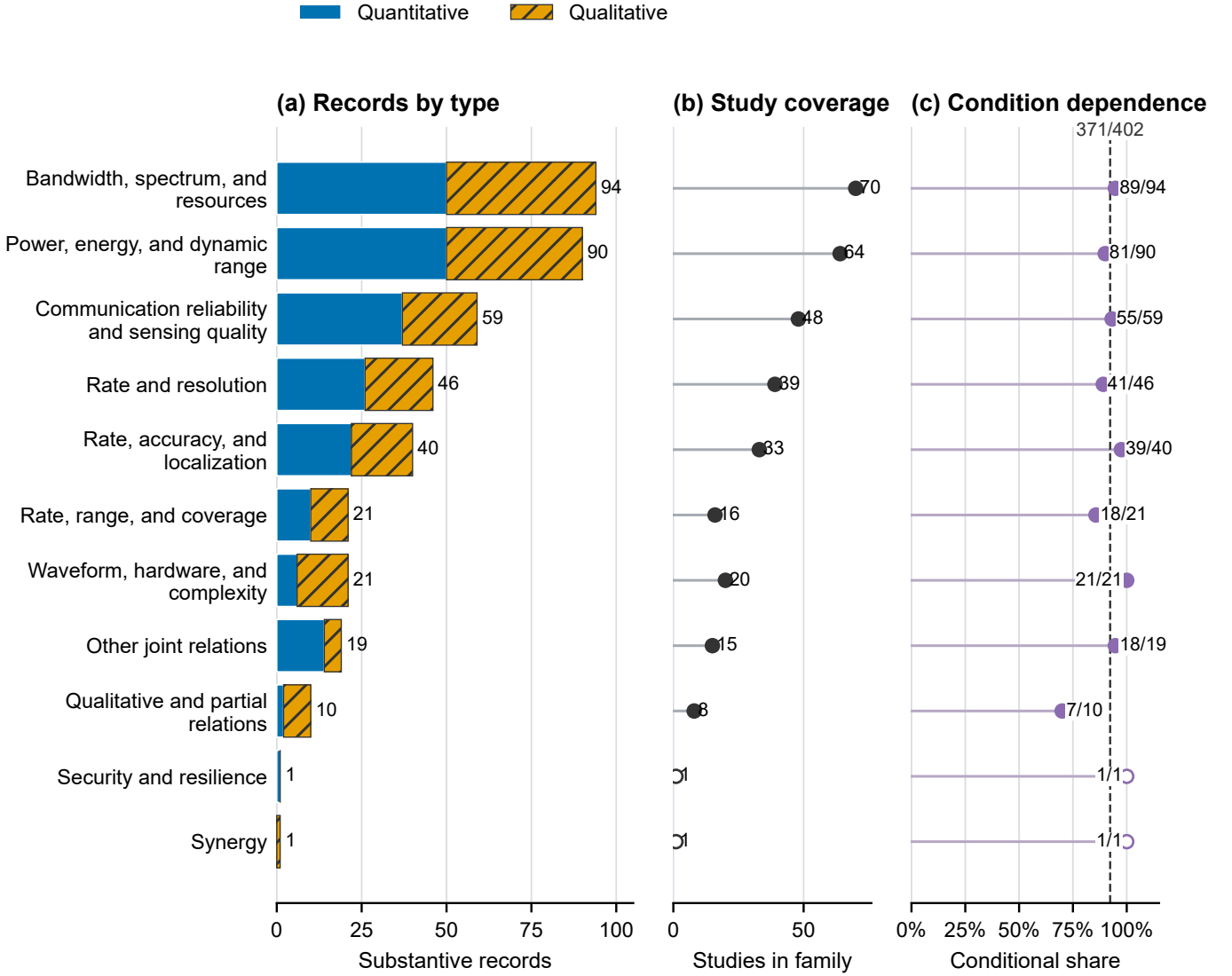


Fig. 8. Evidence profile for 402 substantive joint performance relationships from 168 studies. Panel (a) separates 218 quantitative and 184 qualitative records. Panel (b) reports overlapping study coverage across the mechanism families. Panel (c) reports the share of records whose interpretation depends on stated conditions, with 371 such relationships across the profile. Family study counts may overlap, and frequency describes corpus coverage.

[118]. These results show that signal bandwidth and receiver processing jointly define the operating point.

Reuse extends this mechanism by carrying sensing content in a shared waveform or an existing transport path [25], [119], [120]. Phase coding, carrier grouping, radio frequency bandwidth, and space and wavelength allocation then set the occupancy and isolation available to each function [51], [121], [122], [123]. The result depends on how the receiver separates the shared content and recovers the underlying sequence.

Hardware and processing resources complete the picture. Reference recovery and coding consume timing or redundancy [65], [124], [125], while emitter bandwidth, source coherence, optical budget, and distributed learning traffic set hardware and network capacity [126], [127], [128], [129]. Receiver count introduces alignment and crosstalk constraints. In the reported prototype, four surrounding receivers were designed and one receiver was operated [130]. Phase control and observation settings affect the recoverable path [9], [131], while modu-

lation, sideband, and time bandwidth controls affect signal recovery [132], [133], [134]. The same nominal allocation can therefore produce a different result when the platform or receiver changes.

2) *Power, Energy, and Dynamic Range*: Power affects both functions through the finite operating range of the transmitter, channel, and receiver. Launch and received power, carrier and probe ratios, device bias, modulation swing, and converter headroom refer to different points in that chain. As power changes, performance follows the component that first approaches a noise, nonlinear, or saturation limit. The reference plane and limiting component are therefore part of the result.

A fiber probe sweep shows this response directly. Raising the sensing to telecom power ratio improves sensing SNR and raises BER, with the strongest link penalty near the probe [7]. In a visible light prototype, one optical branch supports communication and ranging while another supports energy harvesting. Their allocation governs rate, ranging quality, and

harvested power under separate receiver and endpoint bias conditions [135]. Similar ratios connect BER or OSNR margin with phase, polarization, and vibration observability in fiber [3], [18], [136], [137], [138]. In photonic wireless links, carrier and waveform ratios connect phase recovery, BER or error vector magnitude (EVM), and radar or ranging quality within the available receiver headroom [32], [139], [140], [141]. Comparisons are most informative when they use the same power reference and receiver margin.

Absolute power sweeps reveal where the operating range ends. In fiber, launch and received power expose nonlinear interaction and sensing reach [21], [142], [143]. Photonic front ends examine operating limits associated with input power, modulator drive, converter linearity, and amplifier gain [38], [144], [145], [146]. Architecture then decides where the available power is used. Polarization paths, receiver taps, and branches assigned to individual functions distribute power before estimation [147], [148], [149]. Propagation, insertion, and alignment losses determine how much reaches those branches [46], [67], [84]. The remaining margin sets the feasible operation of optical phased arrays, communication users, sensing estimators, and illumination services [68], [90], [150], [151].

Power allocation can also be written into the waveform. Service branches, reference copies, and relative waveform amplitudes connect radar masking, ranging, data delivery, and energy harvesting through a shared allocation [152], [153], [154], [155]. Sequence transitions, coding choices, and chaotic source parameters change how communication and sensing information is recovered [156], [157], [158]. Frequency separation, probabilistic shaping, sensing slots, and carrier reuse move energy and occupancy across signal dimensions [159], [160], [161], [162]. The waveform definition, hardware margin, and performance threshold determine the resulting operating point. These same factors also govern reliability and sensing quality.

3) *Communication Reliability and Sensing Quality*: Communication reliability and sensing quality become linked after the waveform and receiver are selected. Communication evidence uses BER, EVM, quality factor, and required OSNR, while sensing evidence uses SNR, estimation error, and detection. These measures can move together as leakage, nonlinear mixing, phase noise, cancellation quality, or receiver competition changes. Some controls reduce an interference path, while others redistribute energy within the shared signal. This distinction explains why similar metric pairs can follow different trends.

Two studies show why interference control and power redistribution require separate interpretation. In a fiber link combining communication with coherent reflectometry, sequence shaping reduces nonlinear interference induced by signal transitions. This lowers the communication penalty while retaining phase sensing at the reported condition [156]. In a millimeter wave waveform, a direct current offset strengthens the sensing component and raises the radar signal to interference ratio together with communication EVM [163]. The first mechanism changes how interference is produced, while the second redistributes energy within the signal. Their

trends therefore arise from different controls.

Architecture then determines where these effects enter the chain. Shared path tests evaluate communication reliability beside sensing outcomes at stated operating points and validation settings [89], [164], [165], [166]. Dedicated monitoring wavelengths and reused channel estimates locate sensing support in optical filtering or receiver processing, with communication performance described qualitatively [33], [167]. The path and receiver therefore define the scope of each reliability result.

Once the architecture is fixed, waveform and inference choices determine the reported communication and sensing metrics. Carrier choice, direct current offset, and phase modulation affect recovery at stated settings [86], [168], [169]. Coding, guard structure, and waveform composition affect BER, correlation quality, and ranging error [170], [171], [172]. At the inference stage, loss weighting connects communication reliability with distance and velocity errors [173], while Kalman tuning acts within the sensing branch [174]. Shared sequence structure and finite block phase processing place radar detection and phase estimation under their sample and noise models [39], [83]. Phase index and pulse position timing likewise connect communication error with radar SNR or time of arrival precision [175], [176]. The sample, noise, timing, and calibration definitions give each result its scope.

At the system level, access placement guided by channel sensing connects link capacity with weather and demand [88]. The aggregation of sensing traffic connects capacity with target count and resolution requirements [41]. Simultaneous use of time and frequency creates direct waveform coupling [177], while shared hardware with waveform switching creates a different path [178]. Waveform robustness under fading and learned phase sensing place the limiting response in the channel or processing stage [179], [180]. These studies connect reliability and sensing quality to the complete waveform, receiver, channel, and processing chain. The same chain also governs rate, resolution, accuracy, and range.

4) *Rate, Resolution, Accuracy, and Range*: The link between communication rate and sensing begins with the design variable that changes the rate. Bandwidth and symbol timing can alter resolution directly, while frame structure, geometry, and processing change the resources available to the estimator. The controlling variable and its operating conditions explain why a similar rate change can produce different sensing outcomes.

Waveform parameters provide the clearest starting point. In optoelectronic oscillator and remote ranging designs, code length and period connect bit rate or capacity with distance resolution and unambiguous range [181], [182]. Power division changes OFDM reliability, reach, and ranging error in a retroreflective prototype, while image decomposition links stripe recovery with scene reconstruction and recognition [19], [183]. LFM slope indexing and time division designs in the W band place their controlling choices in slope selection and frame duration [184], [185]. The shared parameter sets the direction of the relationship in each case.

Frame structure creates a second path by deciding when and where each function receives signal support. Dynamic allocation and intensity scaling distribute support across payload,

sensing patterns, and coverage [4], [186]. Orthogonal time frequency space parameterization and complete grid OFDM reuse place both functions in a common time frequency structure [74], [187]. Separate measurement paths and mode switching show how a similar waveform can support different forms of integration [188], [189]. The frame and measurement schedule are therefore part of the result.

Geometry sets the region in which those resources remain useful. Steering and access point placement define communication and localization coverage, while surface allocation and blockage aware access select the paths available to each function [5], [78], [190], [191]. Source directivity and tag delay spacing further affect coverage, range separation, and concurrent return recovery [192], [193]. The reported layout is therefore part of the observed trend.

Signal construction can tie payload more closely to sensing observability. Chip length changes occupied symbol rate together with ranging precision and security tolerance, while embedded communication data affects distributed sensing fidelity [42], [194]. Amplitude shift keying rate and guard duration connect communication timing with range resolution and ambiguity [195], [196]. Waveform switching provides a related parameter coupling, with rate and resolution evaluated through separate parameter changes [197]. In the terahertz system, the available device band supports quadrature phase shift keying (QPSK) communication and the LFM bandwidth used to achieve range resolution [198]. The waveform and hardware band together define the feasible region.

Optimization studies express these mechanisms as operating regions. Energy and asymptotic block length define communication and detection rate pairs, while nonorthogonal multiple access power allocation connects aggregate rate with ranging error [50], [199]. Other studies combine capacity with ranging information or compare separate and joint operating configurations [77], [200]. Probabilistic shaping and embedded data create further regions through information rate, sidelobe level, and estimation error [201], [202]. The selected operating point follows the stated objective, constraints, and normalization.

Processing adds another layer because the estimator decides how much of the available signal becomes useful sensing information. Sampling period, spectral weighting, zero padding, and transform length exchange response time and computation for resolution under stated acquisition settings [203], [204], [205], [206]. Rate selection can tune the sensitivity of a density estimate derived from BER [207]. In a separate application, classifier size changes recognition accuracy and edge processing burden [62]. These studies distinguish waveform rate from the update and computation needed to interpret it.

Across the network, the relationship moves from a single waveform to the traffic and spectrum available across the system. Transport capacity controls how many sensing streams meet an echo quality threshold [208]. Separated bands, idle time frequency cells, cyclic prefix correlation, and shared optical waveforms connect occupied spectrum with resolution or estimation support [209], [210], [211], [212]. A visible light study carries this logic into a proposed capacity and localization problem [213]. The evidence thus distinguishes observed operation from a proposed design space.

5) *Waveform, Hardware, and Complexity*: Implementation complexity depends on where a design places its burden. Source coherence, receiver structure, calibration, synchronization, computation, memory, and latency represent different costs. Reusing a component may move work into processing or control, so implementation complexity must be assessed across the complete chain.

Two photonic systems show how this transfer occurs. A terahertz design uses coherent comb generation to stabilize the QPSK and LFM paths and concentrates complexity in the optical source chain [198]. A fiber experiment over a long span uses lower cost lasers with broad linewidths and recovers coherent data and vibration through residual carrier processing [214]. Each architecture preserves performance through a different balance of source hardware and digital processing.

6) *Other Implementation Relationships*: Other implementation relationships show how a shared choice appears at different layers of the system. Its effect may emerge as protection, joint gain, device sensitivity, estimator burden, control freshness, or allocation latency. Each case carries its own conditions and prepares the comparison limits that follow.

Security and synergy illustrate two possible directions. Phase keyed recovery uses ranging tolerance to create a physically bounded decryption region [194]. With spatial alignment, communication gain and target profile estimation can benefit from the same optical design [68]. Their geometry, objective, and validation setting explain why one relationship expresses protection and the other expresses joint gain.

Other cases locate the relationship within the implementation chain. Electrode transparency connects device architecture with photodetector sensitivity, while passive return loss and repeater structure affect access for fault localization [60], [215]. Capacity and distance estimation define a model operating region [216]. Queue freshness enters the robotic control loop, and local spectral scanning connects computation with range resolution [47], [217]. Weighted priorities guide resource allocation, while orthogonal task channels shape application latency [218], [219]. Read together, these cases locate the relationship at the device, estimator, control, or allocation layer and provide the conditions needed for comparison across studies.

D. Conditions for Comparison Across Studies

Comparison across studies begins with alignment. The bounded relations summarized in Table V connect studies when their task, metric definition, measurement plane, operating point, and validation setting can be matched through the framework in Section II. Other metric evidence explains mechanisms, feasibility, and behavior within individual systems. This separation keeps each reported value attached to its original engineering meaning.

Tradeoff evidence follows the same principle. Because objectives and constraints vary across systems, the synthesis compares the shared mechanism and the conditions that produce the observed response. This approach shows why a relationship recurs and when it can guide another design. Its

engineering value also depends on the validation setting and on how completely the configuration can be reconstructed. Section VI examines these requirements through validation settings, field evidence across both functional domains, and reconstructability.

VI. VALIDATION EVIDENCE, RECONSTRUCTABILITY, AND BENCHMARK READINESS

Section V showed that comparison depends on alignment across the task, measurement plane, operating point, and validation setting. This section examines how far those operating points were tested and how completely their configurations can be reconstructed. It then relates validation settings, evidence from both functions, and artifact access to benchmark readiness.

A. Validation Settings and Methods

Figure 9 places each study's highest reported setting beside the validation methods used across the corpus. The first panel assigns one setting to every study, while the second allows several methods to contribute to one evidence chain.

The profile shows a broad laboratory and prototype base together with a smaller field subset. Analytical, simulation, laboratory, and prototype methods can contribute to one study, so the configuration trail across methods shapes interpretation.

These settings expose different parts of an O-ISAC system. Simulation studies isolate waveform and device sensitivities under controlled conditions [85], [187]. Laboratory and prototype systems reveal conversion, synchronization, and receiver effects in physical implementations [60], [74], [110]. Field studies extend the evidence to access networks, deployed fiber, submarine monitoring, and outdoor wireless links [20], [33], [37], [38], [123].

The highest setting describes the reach of the evidence. Its scientific meaning still depends on the claim and the test design. A controlled prototype may explore a wider parameter range than a field trial, while a laboratory experiment may reveal communication and sensing interactions more directly than a field dataset collected for one outcome. The setting therefore gains meaning when it is read together with the tested conditions, available controls, and functional coverage.

Each method serves a different analytical purpose. Analysis clarifies assumptions, simulation supports controlled parameter studies, and laboratory work reveals the behavior of physical components. Prototypes bring subsystems together, while field studies introduce at least one operational boundary. Their combined value grows when the configuration and conditions remain traceable as the evidence moves from one method to another.

Section IV shows that realistic validation conditions change with the platform. Fiber studies depend on topology, traffic, launch and receiver conditions, and the disturbance context. Visible light communication and LiFi studies add layout, orientation, illumination, detector, and processing conditions. Free space optical studies require geometry, pointing, target properties, weather, and motion. Terahertz systems enabled by

photonics must also trace optical conversion, antenna alignment, and wireless propagation. A useful validation record therefore identifies the environment, measurement plane, target or disturbance, reference method, and exercised function.

B. Field Evidence Across Communication and Sensing

The highest validation setting and the coverage of both functions answer related questions. For 12 studies, a field trial or deployment was the strongest reported setting. Six of them also received TQAF validation score 3 because their records report field or deployment outcomes in each functional domain. The assessment is applied to each study. Electronic Supplement S-Appraisal reports the corresponding scores. Electronic Supplement S7, the paired function validation register, identifies the 12 studies and the subset of six.

The six studies cover several settings. They include terahertz and cellular links enabled by photonics, fiber access integrated with radio frequency identification, and space laser ranging and communication. They also include multicore fiber deployed in the field and submarine cable monitoring [20], [33], [37], [38], [65], [123]. Their diversity shows that field evidence for both functional domains appears across several platforms. Platform comparisons still follow the native validation conditions described in Section IV.

Electronic Supplement S7 records the available relationship timing and separate source locators for the two functions. The six cases establish field or deployment outcomes in both functional domains at the level of each study. Concurrent joint operation is established when both outcomes are traceable to the same configuration, time interval, and condition set.

The distinction becomes important when evidence is produced at different stages of a system. One fiber study combines field data with offline sensing classification and illustrates how field collection and functional execution can occupy different parts of the evaluation path [120]. Segmented architectures add interfaces, timing, calibration, and conversion conditions to that path. When these elements connect both outcomes to one operating point, field coverage can support a stronger claim of joint operation.

C. Artifact Access and Reconstructability

Validation describes what a study tested. Reconstructability describes how completely another group can recover the operating point and the path from input to reported result. Access to data, code, or models supports this task, while physical experiments also depend on the system configuration, measurement conditions, processing steps, and provenance.

Table VI summarizes the access routes reported for data and for code or models in all 206 studies.

The reported routes support different forms of reconstruction. Open data appear across simulation, laboratory, prototype, computer vision, and field data settings [91], [110], [120], [187], [220]. They support reanalysis of the released data. Complete system reconstruction also requires configuration and execution details. Software preserves the transformation from inputs to reported outputs, and one computer vision study reported partial code or model components [91].

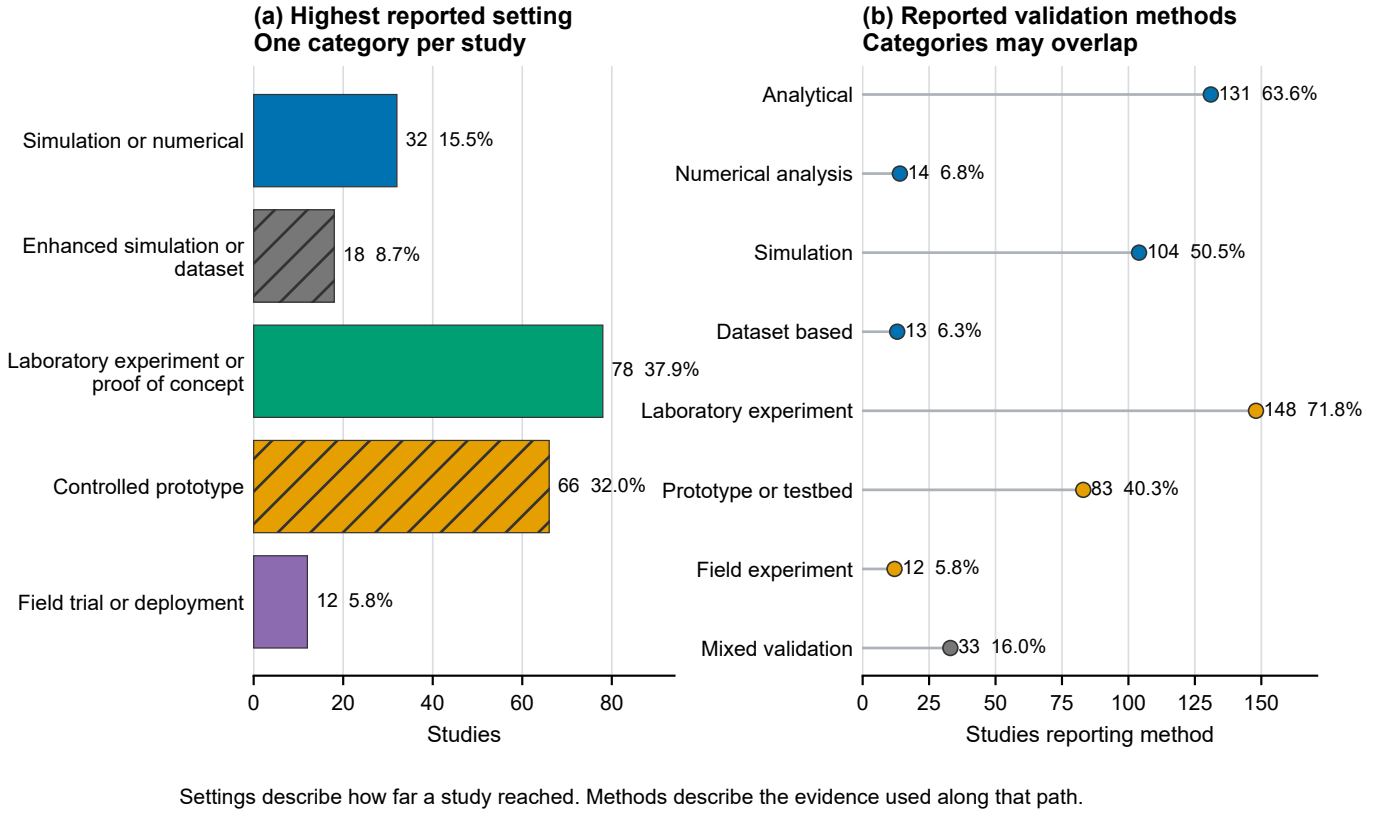


Fig. 9. Validation settings and methods in the 206 studies. Panel (a) assigns each study to one highest reported setting. Panel (b) reports analytical and numerical methods, simulations, methods based on datasets, laboratory experiments, prototypes, field studies, and mixed methods. Method categories may overlap because several forms of evidence can contribute to one study. The panels describe the reach and composition of validation as distinct properties.

TABLE VI
REPORTED ACCESS TO DATA AND TO CODE OR MODELS IN THE 206 STUDIES

Reported access	Studies, <i>n</i> (%)
<i>Data</i>	
Closed or status unreported	145 (70.4%)
Access by request	41 (19.9%)
Open access	13 (6.3%)
Outside artifact scope	7 (3.4%)
<i>Code or model</i>	
Closed or status unreported	197 (95.6%)
Access by request	7 (3.4%)
Partial components	1 (0.5%)
Outside artifact scope	1 (0.5%)

Note. The first row in each group combines closed access and an unspecified access status.

Reconstructing a physical experiment requires the system and hardware, the scene and calibration, the waveform and processing, the metrics and data, and the versioned release. These requirements extend the comparison profile in Section II. Electronic Supplements S-Data Dictionary and S-Evidence provide the field definitions and record provenance used in the review.

The TQAF reproducibility assessment reads artifact access together with parameter reporting. It places 4 studies in the low category, 199 in the adequate category, and 3 in the strong category. Electronic Supplement S-Appraisal provides scores

for each study. Thirteen studies report open data, while three reach strong reproducibility. The difference reflects the additional reconstruction information captured by the appraisal.

D. Benchmark Readiness

A reconstructable result becomes a reusable benchmark when another group can apply the same task, condition set, baseline, and metric definition. The TQAF benchmark assessment combines these requirements with artifact access, validation, reproducibility, and comparison admissibility. The observed distribution comprises 48 studies in the low category, 158 in the adequate category, and zero in the strong category.

Under the review criteria, strong readiness depends on bringing an external or common baseline, an open artifact, admissible outcomes in both functional domains, and strong validation and reproducibility support into the same evidence chain. S-Appraisal reports these scores. This distribution directs attention toward shared tasks, baselines, artifacts, and evaluation conditions that other groups can reuse.

These distributions characterize evidence reach, functional coverage, artifact access, and reconstructability within the 206 included studies. They provide the descriptive basis for the benchmark and reporting actions in Section VIII.

Overall, the corpus combines a broad laboratory and prototype base with a smaller set of field and deployment results. Stronger cumulative evidence emerges when both functions are tied to one traceable configuration and the evaluation path is

preserved through reusable artifacts. Section VII builds on this validation map by examining the technologies and applications supported by the reported evidence and their connection to 6G network roles.

VII. ENABLING TECHNOLOGIES, APPLICATION REQUIREMENTS, AND 6G NETWORK EVIDENCE

Validation shows how far a reported system was tested. We now follow the technologies that form communication and sensing observables into application requirements and 6G network roles. Because a study may contribute to several technology and application labels, their counts describe coverage within the corpus.

Figure 10 follows this evidence chain from optical generation and transport through waveform formation, spatial control, inference, selected application domains, and 6G network positioning.

The chain shows that network interpretation depends on continuity across the physical observable, the joint design choice, the application condition, and the reported network role. The displayed categories guide the reader through that relationship, while the following subsections retain the platform conditions and study evidence.

A. Technologies That Create and Shape Observables

The main technology labels occupy complementary points in the signal chain. Photonic generation creates or translates the carrier. Chirped processing maps delay or motion to beat frequency. Coherent reception preserves phase, and OFDM provides controllable time and frequency resources. A single architecture can combine several of these mechanisms.

OFDM supports joint operation by placing payload and sensing resources within one signal structure. The sensing observable may come from pilots, payload tones, cyclic prefixes, reserved subcarriers, or a dedicated processing branch. This choice determines which samples are known at the receiver and how interference enters the estimate. Optical modulation constraints and the measurement plane used for SNR shape the interpretation. Embedded pilot fiber sensing and indoor optical positioning show how the same multicarrier principle serves different observables and geometries [85], [192].

Chirped and coherent systems draw on phase continuity and a shared timing reference. A returned chirp can reveal range or motion as the transmitted signal carries data. A coherent front end preserves small phase changes for distributed sensing and demodulation. Photonic generation can extend the reference to wideband millimeter wave or terahertz carriers. Full photonic dechirping and coherent fiber vibration sensing illustrate these two uses [42], [198]. Their performance depends on laser stability, reference distribution, conversion efficiency, receiver bandwidth, and calibration.

The optical stage governs carrier generation and reference stability. Once the carrier is radiated, the communication and sensing paths follow microwave, millimeter wave, or terahertz propagation. In a free space optical link, the optical beam itself traverses the channel. System evaluation should therefore follow the signal across both physical domains and identify

where power, SNR, bandwidth, and timing are measured. This view makes the interface between components part of the joint system.

Spatial control changes both the communication link budget and the sensing geometry. Beamforming, optical phased arrays, multiple input multiple output processing, and reconfigurable optical intelligent surfaces provide different forms of spatial control. Beamforming directs energy. An optical phased array provides one steering mechanism. Multiple input multiple output processing uses several spatial dimensions, and an intelligent surface reshapes an intermediate propagation boundary. A system may combine these roles.

A narrow beam improves received power and angular selectivity while making acquisition, tracking, blockage, and calibration central to joint performance. The joint use of an optical phased array and OFDM and the use of frequency comb beamforming connect steering with ranging or imaging [68], [110]. Multiple input multiple output processing adds communication streams, sensing diversity, or separation among targets and paths. Structured fields and hybrid systems that combine free space optics with fiber Bragg gratings place these spatial dimensions at different points in the architecture [76], [81]. Meaningful comparison therefore requires the independent dimensions, geometry, and calibration method.

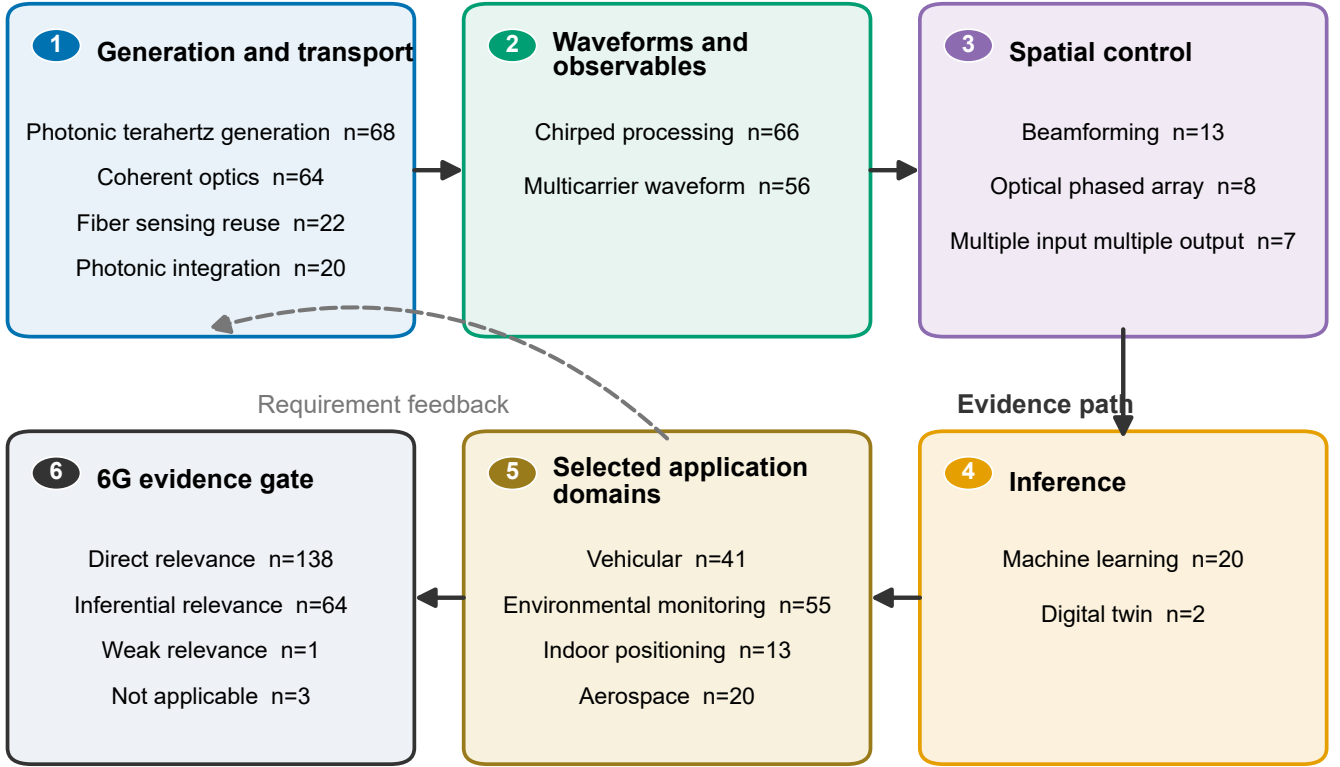
The intelligent surface evidence consists of one study that models a mobile free space optical path controlled by an optical intelligent surface [191]. A separate laboratory demonstration uses mechanically rotated Risley prisms to steer a photonic generated terahertz signal [109]. Both mechanisms redirect energy, and each uses different surface physics, propagation laws, control variables, and validation settings. Their separate treatment preserves the physical meaning and provides two starting points for broader experimental comparison.

B. Integration Beyond the Waveform

Integration and inference move joint design beyond waveform selection. Photonic integration appears in 20 studies, fiber distributed acoustic sensing in 22, and machine learning or artificial intelligence in 20. Two studies use digital twins. A further 19 studies contain specialized mechanisms grouped as other because they fall outside the broad technology labels. Together, these categories cover device integration, infrastructure reuse, inference, and a diverse long tail.

Photonic integration can place generation, reference distribution, beam control, and detection on a compact hardware path. This arrangement reduces duplicated paths and brings packaging, thermal drift, crosstalk, and calibration into the same design problem. Microcomb synchronization coordinates wireless data with radar imaging. A distributed W band system uses radar estimates for beam alignment and flexible data service [221], [222]. Both examples show how a shared reference can connect communication and sensing hardware.

Fiber distributed acoustic sensing follows a complementary route by using the transmission medium as a spatial sensor. Phase, backscatter, and polarization can reveal events along the route as the fiber carries traffic. Coherent multicore transmission with distributed sensing and operational submarine



Technology and application labels may overlap. The 6G evidence gate assigns one class per study.

Fig. 10. Organization across system layers of the mechanisms that create and shape O-ISAC observables and connect them to service requirements. Technology and application counts describe overlapping corpus coverage, so a study may contribute to several displayed categories. The 6G evidence gate is exclusive and assigns each of the 206 studies to one relevance class. Selected application domains illustrate the requirement flow, while Table VII provides the complete application inventory.

cable monitoring illustrate two scales of this infrastructure reuse [33], [147]. The central design question becomes how to preserve the optical observable together with traffic continuity.

Learning turns optical observations into event labels, positions, field estimates, channel predictions, and resource decisions. Each task is best read with its definition, reference data, training distribution, evaluation shift, uncertainty, and inference cost. Fiber diagnosis, federated visible light positioning, and wearable optical sensing show the range from distributed infrastructure to personal use [61], [120], [129].

A neural network can also take part in both functions. In one coherent link, the model compensates transmission impairments and estimates distributed power and abnormal loss [223]. The two outputs share the link while retaining separate targets and reference data. Digital twins add another model layer. One represents vehicular edge resources, and another provides a model aided physical reference for coherent fiber sensing [180], [219]. Clear reporting of the represented state, update process, mismatch model, and resulting decision gives these models an explicit system role.

C. Applications as Operating Requirements

Applications translate the mechanisms above into conditions that a joint service must satisfy. Table VII retains all thirteen application domains and maps each one to its joint role and defining evaluation conditions. A fiber platform for vehicles, for example, carries data at high rates and measures distributed temperature [224]. Devices used by people connect communication or remote alerts with motion, falls, air pressure, physiology, or gesture [61], [225], [226]. Each application therefore changes what the experiment must demonstrate, even when both functions share the same optical interface.

Across the map, the dominant requirement moves with the application. Network and mobility settings emphasize acquisition, timing, and continuity. Deployed infrastructure emphasizes persistent observation alongside carried traffic. Human use and extreme settings add physiological references, safety, privacy, pointing, atmosphere, packaging, and access to the test environment. These requirement bundles show which operating conditions must accompany a reported result before it can support comparison or reuse.

TABLE VII
APPLICATION DOMAINS AND OPERATING REQUIREMENTS IN THE REVIEWED CORPUS

<i>Network and mobility</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
6G and access 100 (48.5%)	Access and transport with channel, target, position, or environment sensing	Timing, orchestration, continuity, and evidence for both functions	[41], [76]
Vehicular 41 (19.9%)	Vehicle data links with range, motion, pose, temperature, or channel sensing	Changing geometry, acquisition, tracking, and service continuity	[49], [224]
Optical access 32 (15.5%)	Access and transport links with channel, event, or fiber state sensing	Reach, synchronization, and traffic continuity	[41], [85], [120]
Indoor positioning 13 (6.3%)	Indoor optical links with user or robot position and orientation sensing	Geometry, calibration, coverage, and update continuity	[47], [192]
Data center 4 (1.9%)	Short reach data links with channel estimation or link state sensing	Traffic continuity and calibrated link references	[85]
<i>Infrastructure</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Environmental monitoring 55 (26.7%)	Distributed links with vibration, temperature, or environmental event sensing	Persistent observation, localization, and traceable reference data	[33], [120], [206]
Smart infrastructure 41 (19.9%)	Infrastructure links with structural, motion, or event monitoring	Continuous observation, calibration, and resilient alert delivery	[76], [120], [126]
Industrial 34 (16.5%)	Industrial data or control with process, strain, temperature, motion, or position sensing	Delay, continuity, calibration, and safe operation	[61], [76]
Security and surveillance 26 (12.6%)	Monitoring links with anomaly, event, or location sensing	Coverage, event discrimination, and alert latency	[120]
<i>Human use, extreme settings, and remaining domains</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Healthcare 6 (2.9%)	Wearable or ambient links with vital sign, activity, or gesture sensing	Motion robustness, physiological reference, illumination, and user safety	[60], [61]
Aerospace 20 (9.7%)	Satellite links with atmospheric or wavefront sensing	Pointing, turbulence, delay, and link adaptation	[73]
Underwater 2 (1.0%)	Underwater link concepts and subsea data links with temperature or event sensing	Water attenuation or cable calibration, packaging, and access	[126], [206]
Other 17 (8.3%)	Emerging links with range, imaging, object, or channel sensing	Metrics and reporting specific to the task	[109]

Note. Application domains overlap, so a study may contribute to more than one row. Counts and percentages describe coverage within the 206 included studies. For each domain, the table includes a selected study set and lists the corresponding references in the final column. The 6G and access row identifies the application category, and the final subsection provides the 6G relevance classification.

D. From 6G Relevance to Network Evidence

The 6G relevance gate in Figure 10 assigns each study one corpus position. Direct cases connect the reported system explicitly to a 6G role, as illustrated by [41], [110]. Inferential cases contribute an enabling link that gains meaning from additional network context, as illustrated by [42], [85]. The remaining classes capture a lighter connection to 6G or an application focus outside that framing, as illustrated by [167], [226].

The distribution connects the reviewed corpus with 6G access, transport, localization, automation, environmental awareness, and generation of carriers at high frequencies. A relevance code therefore locates a study within the 6G discussion. Standard conformance, interoperability, and deployment readiness require additional evidence tied to the relevant network framework [2].

A stronger network interpretation emerges when the optical or photonic subsystem occupies an explicit role across the complete network path. Access, fronthaul, transport, edge processing, and service control bring timing, mobility, orchestration, energy, security, and recovery requirements. Evidence at this level evaluates useful communication and sensing under

the conditions of that role and reports the resources needed to sustain both. This creates a traceable path from the physical observable and hardware to the service objective.

Within this survey, 6G relevance is strongest when both functional outcomes and their operating conditions remain traceable to a network role. The resulting evidence chain links physical observables, application requirements, and service objectives and provides the basis for the research priorities in Section VIII.

VIII. DISCUSSION, RESEARCH ROADMAP, AND LIMITATIONS

Across the evidence base, the central issue is whether physical sharing produces joint outcomes that remain traceable to one operating context. This section interprets that relationship, develops five research priorities, and states the review boundaries.

A. Interpretation Across the Evidence Base

Physical integration identifies where communication and sensing share infrastructure, signals, or information. Evidential integration shows how clearly both outcomes remain tied to the

same operating conditions. Together, they connect architectural and transmission perspectives while preserving the physical regime of each platform [11], [12], [14], [16].

Deeper sharing can connect more design variables and improve coordination across the two functions. It can also couple power, distortion, synchronization, dynamic range, and failure behavior. Its scientific value therefore depends on locating these effects in the signal chain and tracing their influence on both outcomes.

Tradeoff patterns transfer across studies when tasks, metric definitions, measurement planes, baselines, constraints, and validation settings remain aligned. This alignment allows model bounds and empirical operating points to retain their distinct roles and support comparison within a stated context [15], [227].

Fielded cable and network studies illustrate the value of operational continuity [228], [229]. Within the survey, system evidence becomes reusable when configuration, calibration, processing, and reported artifacts remain connected across interfaces. This continuity links physical sharing to application and network claims and motivates a central research task. Joint operating points should remain interpretable, testable, and reusable across groups.

B. Research Priorities for Reusable Evidence

Table VIII organizes five priorities and pairs each observed finding with an action and a measurable sign of progress.

The order matters because each stage uses evidence produced by the previous one. Clear measurement context supports fair benchmarks. Benchmarks support realistic joint tests. Reconstructable artifacts make those tests repeatable, and repeatable tests provide a basis for adaptive and secure evaluation at network scale.

Stress tests should follow the native signal path of each modality. Fiber tests can vary topology and traffic, free space optical tests can vary geometry and atmosphere, and visible light tests can vary illumination and orientation. Photonic terahertz tests can vary optical drift together with wireless propagation. Matched baselines for the individual functions keep the limiting mechanism visible across these settings.

For standards and 6G coordination, the same sequence can support a core reporting profile that preserves modality physics, ground truth, and reusable benchmark definitions. Shared metadata can connect results across platforms while retaining the conditions needed for interpretation.

C. Scope and Limitations of the Review

The review represents English language technical reports published from 1 January 2020 through 22 June 2026 and indexed by the six searched sources. Full texts were available and assessed for 272 of the 330 reports sought, which corresponds to 82.4% retrieval coverage. Two Taylor & Francis exports with few records are documented through their complete platform exports. These boundaries define the literature represented by the final corpus of 206 studies.

Searches and screening preceded retrospective registration, and the subsequent search refresh and report reconciliation

TABLE VIII
RESEARCH PRIORITIES DERIVED FROM THE REVIEWED EVIDENCE

Priority	Finding and proposed action	Evidence of progress
1. Measurement contract	<i>Finding.</i> Of 4,779 metric records, 118 supported bounded relations under aligned conditions. <i>Action.</i> Record the task, plane, unit, baseline, operating condition, validation setting, and source locator.	An independent reader reconstructs the quantity and condition from a structured record.
2. Benchmark for each modality	<i>Finding.</i> Comparison across studies depends on matched physical context. <i>Action.</i> Repeat one task with reference and independent designs under shared calibration and ground truth.	Independent groups reproduce the relative finding under the shared protocol.
3. Joint operation under disturbance	<i>Finding.</i> Of 402 substantive tradeoff relationships, 371 depend on reported conditions. Six of 12 field or deployment tier studies report outcomes in both domains. Traceability is strongest when both outcomes share the same timing and condition set. <i>Action.</i> Evaluate both functions in the same run and condition set under load, a disturbance relevant to the modality, and matched baselines for the individual functions.	Repeated trials recover the joint region and its limiting mechanism.
4. Reconstructable experiment package	<i>Finding.</i> Open data appear in 13 of 206 studies, and code or model access reaches a partial release in one. <i>Action.</i> Preserve a versioned package and rerun the main comparison under one added condition.	An independent rerun recovers the comparison and added condition from the package.
5. Intelligence, scale, and security	<i>Finding.</i> Technology and application coverage connects the corpus to 6G roles. Conformance, interoperability, and readiness become measurable through explicit network tests. <i>Action.</i> Apply a declared shift, attack, load, or component failure and report overhead, resource use, decisions, and fallback.	The service retains an explainable boundary with resource use, decisions, and fallback recorded.

Note. The findings summarize the retained corpus presented in Sections IV through VII. The proposed actions retain the physical conditions of each modality and define observable signs of progress.

produced the final corpus described in Section III. One investigator supervised routine decisions, while independent checks focused on specified evidence records and report reconciliation.

TQAF provides a technical appraisal developed for this review. The diversity of tasks, quantities, validation settings, outcomes, and source protocols supports structured narrative synthesis, which preserves native conditions and bounded comparisons. This analytical scale centers interpretation on reported mechanisms and conditions. Harmonized outcomes and source protocols form the appropriate basis for pooled effects, sensitivity estimates, and causal moderator analysis.

The multilabel taxonomy incorporates reviewer judgment, especially for hybrid systems. Row counts represent coded observations at extraction granularity, while effect analysis uses a different unit. For 92 metric rows, the available fields supported interpretation within the source setting and a con-

servative comparison status. Artifact access combined source reporting with targeted link checks, so the appraisal reflects documented reconstruction information.

The conclusions concern recurring mechanisms, measurement constraints, and validation patterns within the retained corpus. Corpus distributions describe the reviewed literature. Worldwide prevalence, market maturity, and platform rank require evidence designed for those questions. Within its stated scope, the review identifies the conditions that allow optical integration to support transferable joint results.

IX. CONCLUSION

Across the reviewed corpus, O-ISAC is organized most clearly by the physical role of optics and the mechanism that couples communication and sensing. Optics may carry the user link, illuminate the target, distribute a carrier, or preserve a coherent reference, and each role creates a different set of constraints. Shared hardware, carriers, waveforms, resources, links, and design rules then connect the two functions at different points in the system. Their engineering value depends on how the task, metric definition, measurement plane, operating condition, and validation setting are reported together.

The evidence indicates that recurring tradeoff mechanisms can be compared when their physical context and measurement contracts align. Laboratory and prototype studies provide a broad experimental base, while field results illustrate how topology, geometry, propagation, calibration, and service continuity enter real systems. Application and 6G relevance become more informative when both functional outcomes remain traceable to the same network role and condition set. This makes transferable joint evidence the central measure of progress and keeps individual performance values tied to their source conditions.

Progress therefore depends on measurement contracts tailored to each modality, paired validation under realistic disturbances, and versioned artifacts that preserve configuration, calibration, data, and processing. These practices provide a path from individual demonstrations to reusable benchmarks and evaluation at network scale. They also allow subsequent designs to build on prior evidence while preserving the physical meaning of each optical platform.

REFERENCES

- [1] D. Zhang *et al.*, "Integrated sensing and communications over the years: An evolution perspective," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 5014–5048, 2026.
- [2] International Telecommunication Union, "Framework and overall objectives of the future development of IMT for 2030 and beyond," ITU Radiocommunication Sector, Tech. Rep. Recommendation ITU-R M.2160-0, Nov. 2023.
- [3] J. Wang, L. Lu, J. Zhang, A. P. T. Lau, Z. Li, and C. Lu, "On the coexistence of distributed fiber optic sensing and IM/DD transmission via digital subcarrier multiplexing," *Optics Express*, vol. 34, no. 3, p. 5592, 2026.
- [4] Z. Li, J. Shi, C. Shen, Y. Zhang, J. Zhang, and N. Chi, "Adaptive visible light integrated sensing and communication for cable-free virtual reality," *Photonics Research*, vol. 13, no. 9, p. 2453, 2025.
- [5] X. Guo, Z. Li, Z. Zhou, L. Deng, Y. Yu, and X. Zhang, "Millimeter-Wave Integrated Sensing and Communication System Based on a Photonics-Assisted Phased Array," *Laser & Photonics Reviews*, vol. 20, no. 10, p. e02350, 2026.
- [6] R. Kaide *et al.*, "Terahertz ISAC with Simultaneous Fast-Swept FMCW Radar and High-Speed Wireless Link Using a Single UTC-PD," *Electronics*, vol. 15, no. 8, p. 1608, 2026.
- [7] L. Maoqi, W. Jingchuan, L. Chen, Y. Changyuan, and L. Chao, "Pre-Compensation of LFM Sensing Probe via Integrated Sensing and Communication over Fiber System," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, 2025, pp. 1–4.
- [8] Y. Chen *et al.*, "Optical Integrated Sensing and Communication Lidar Based on RBPSK-FMCW Joint Waveform and Optical Phased Array," *Journal of Lightwave Technology*, vol. 44, no. 10, pp. 4032–4040, 2026.
- [9] M. Zhou, R. Jin, S. Chen, J. Ding, M. Xu, and Z. Wu, "Turbulence-Resilient Communication through Beam Shaping Enabled by Multi-Region Turbulence Intensity Sensing," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–5.
- [10] S. A. H. Mohsan *et al.*, "Optical integrated sensing and communication: Fundamentals, applications, challenges and future aspects," *Optical Switching and Networking*, vol. 61, p. 100854, 2026.
- [11] Y. Wen, F. Yang, J. Song, and Z. Han, "Optical integrated sensing and communication: Architectures, potentials and challenges," *IEEE Internet of Things Magazine*, vol. 7, no. 4, pp. 68–74, 2024.
- [12] C. Liang *et al.*, "Integrated sensing, lighting and communication based on visible light communication: A review," *Digital Signal Processing*, vol. 145, p. 104340, 2024.
- [13] Z. Lyu *et al.*, "Exploring photonic THz-ISAC systems with integrated waveforms," *Journal of Lightwave Technology*, vol. 44, no. 3, pp. 868–877, 2026.
- [14] X. Su *et al.*, "Optical integrated sensing and communication: From theory to practice," *Tsinghua Science and Technology*, 2025, available online 11 December 2025.
- [15] A. G. Khorasgani, "A survey on general and optical ISAC: Fundamental limits, design approaches, and future directions," *Optical Switching and Networking*, vol. 60, p. 100853, 2026.
- [16] Z. Liu *et al.*, "A review of optical fiber integrated sensing and communication systems," *Science China Information Sciences*, vol. 69, no. 6, p. 160302, 2026.
- [17] L. Lu, Y. Yan, B. Yang, Y. Yang, A. P. T. Lau, and C. Lu, "Towards converged networks: A review of integrated optical fiber sensing and communication," *Science China Information Sciences*, vol. 69, no. 6, p. 160301, 2026.
- [18] L. Wang *et al.*, "Integrated Communication and In-Band Spectrum Polarization-Based Sensing via Fraction-Division Non-Orthogonal Multiple Access," *Journal of Lightwave Technology*, vol. 44, no. 1, pp. 138–147, 2026.
- [19] J. Du *et al.*, "Flexible Waveform Design for RO-ISAC Based on OFDM-Embedded Maximum Length Sequence," *IEEE Wireless Communications Letters*, vol. 14, no. 10, pp. 3334–3338, 2025.
- [20] Q. Wang *et al.*, "Real-Time Integrated Sensing with 5.08 mm Range Resolution and 15.5 Gbps Bidirectional Wireless Communication Over a 270-Meter Outdoor Photonic-Assisted Terahertz Link," *IEEE Photonics Journal*, vol. 18, no. 4, p. 7301309, 2026.
- [21] X. Tang *et al.*, "Optical fiber communication and distributed multi-parameter sensing integrated network for smart power grid applications," *Science China Information Sciences*, vol. 69, no. 6, p. 160305, 2026.
- [22] M. J. Page *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, 2021.
- [23] M. L. Rethlefsen *et al.*, "PRISMA-S: An extension to the PRISMA statement for reporting literature searches in systematic reviews," *Systematic Reviews*, vol. 10, no. 1, p. 39, 2021.
- [24] Z. Lyu *et al.*, "Multi-Channel Photonic THz-ISAC System Based on Integrated LFM-QAM Waveform," *Journal of Lightwave Technology*, vol. 42, no. 11, pp. 3981–3988, 2024.
- [25] J. He, S. Xiao, C. Ying, Z. Chen, and Y. Xiao, "Sliding Window-Based LSTM Scheme for PAM-4 Photonics-Assisted W-Band ISAC Systems," *IEEE Photonics Technology Letters*, 2026.
- [26] D. Shi *et al.*, "310Gbps Transmission and Centimeter-Level Ranging Photonic THz ISAC System," *Journal of Lightwave Technology*, 2026.
- [27] S. Ye *et al.*, "Photonic Frequency-Hopped THz ISAC-Enabled by Electro-Optically Tunable Laser," *IEEE Photonics Technology Letters*, 2026.
- [28] R. Kaide, S. Ye, Y. Wang, Y. Mikami, Y. Ueda, and K. Kato, "Integrated Terahertz FMCW Radar and FSK Communication Enabled by High-Speed Wavelength Tunable Lasers," *Photonics*, vol. 12, no. 10, p. 977, 2025.

- [29] R. Wang, W. Ma, J. Liu, H. Xu, Z. Shen, and W. Li, "Integrated sensing and communication system based on a photonic integrated broadband multi-beamformer," *Optics Express*, vol. 33, no. 10, p. 20619, 2025.
- [30] Y. Wang, W. Lu, Y. Song, and J. Yu, "Photonics-Assisted mmWave Integrated Sensing and Communication System Enabled by ODDM Waveform," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [31] T. Reddy, V. Reddy, R. Thipparaju, and U. Singh, "Performance Analysis of Microwave Photonics Assisted Joint Radar Communications," in *2024 International Conference on Wireless Communications, Signal Processing and Networking, WiSPNET 2024*, 2024.
- [32] B. Dong *et al.*, "Photonic-Based W-Band Integrated Sensing and Communication System With Flexible Time-Frequency Division Multiplexed Waveforms for Fiber-Wireless Network," *Journal of Lightwave Technology*, 2024.
- [33] S. Chen, K. Zhu, J. Han, Q. Sui, and Z. Li, "Photonic Integrated Sensing and Communication System Harnessing Submarine Fiber Optic Cables for Coastal Event Monitoring," *IEEE Communications Magazine*, vol. 60, no. 12, pp. 110–116, 2022.
- [34] H. Zhou *et al.*, "Ultrasonic phase extraction method for co-cable identification in coherent optical transmission systems," *Chinese Optics Letters*, 2024.
- [35] S. Gaiani, M. Fasano, A. Gatto, P. Parolari, P. Martelli, and P. Boffi, "Exploitation of FMF capabilities for joint communications and sensing," in *International Conference on Transparent Optical Networks*, 2025.
- [36] C. Guan *et al.*, "A novel passive bi-directional audio over fiber transmission system with 20 Km," *Optical Fiber Technology*, 2024.
- [37] S. Zhang, J. Li, J. Zhu, D. Zhang, Y. Zeng, and Y. Luo, "First Demonstration of FTTR-based RFID Network for Integrated Sensing and Communication," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–3.
- [38] Y. Zhao, J. Wan, S. Xu, and W. Zou, "Integration of endogenous sensing and communication based on photonics-enabled cellular base station networks with a spectrum fusion algorithm," *Optics Express*, vol. 32, no. 26, p. 47428, 2024.
- [39] A. Alzamil, M. Sharawy, E. Almohimmah, A. Ragheb, A. Almainan, and S. Alshebeili, "Development of an Integrated Communication and Sensing System Using Spread Spectrum and Photonics Technologies," *Photonics*, 2024.
- [40] M. Lei *et al.*, "A Spectrum-Efficient MoF Architecture for Joint Sensing and Communication in B5G Based on Polarization Interleaving and Polarization-Insensitive Filtering," *Journal of Lightwave Technology*, 2022.
- [41] A. Tzanakaki and M. Anastasopoulos, "Optical Transport Networks Supporting Integrated Communications and Sensing in 6G," in *2025 European Conference on Optical Communications (ECOC)*, 2025, pp. 1–4.
- [42] H. He *et al.*, "Integrated sensing and communication in an optical fibre," *Light: Science & Applications*, vol. 12, no. 1, p. 25, 2023.
- [43] Z. Hu, C. Zhao, M. Zhang, J. Chen, and M. Tang, "Native and Reconfigurable Distributed Acoustic Sensing Integrated with Single-carrier 64QAM Transmission Signal," in *Optical Fiber Communication Conference (OFC 2025)*, 2025.
- [44] H. Zhu, X. Xu, Z. Liu, and J. Zhang, "Integrated optical covert sensing and communication," *Chinese Optics Letters*, 2025.
- [45] L. Gu *et al.*, "Co-wavelength-channel integration of ultra-low-frequency distributed acoustic sensing and high-capacity communication," *Photonics Research*, vol. 13, no. 6, p. 1611, 2025.
- [46] P. J. Urban, E. M. Miśkiewicz, D. Karczewicz, W. Kogut, and J. Koczyński, "Parallel Path Effect-Free Remote Monitoring and Sensing in Live XGS-PON Using Legacy Optical Time Domain Reflectometry," *Journal of Lightwave Technology*, vol. 43, no. 14, pp. 6711–6719, 2025.
- [47] S. Wang and H. Chen, "Optical Integrated Sensing and Communication for Cooperative Mobile Robotics: Design and Experiments," in *2023 21st International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, 2023, pp. 207–214.
- [48] Y. Song *et al.*, "VehicleTalk: Lightweight V2V Network Enabled by Optical Wireless Communication and Sensing," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, 2024, pp. 1–5.
- [49] K. Dong, W. Cao, and M. Wang, "Optical Camera-Based Integrated Sensing and Communication for V2X Applications: Model and Optimization," *Sensors*, vol. 25, no. 22, p. 7061, 2025.
- [50] N. N. Luong, C. T. Nguyen, and T. V. Pham, "Performance Analysis of NOMA-Assisted Optical OFDM ISAC Systems with Clipping Distortion," in *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- [51] Z. Liu, J. Zhu, J. Yang, Y. Sun, Y. Yang, and J. Luo, "ReflexGest: Recognizing Hand Gestures Under VLC-Capable Lamps," *IEEE Transactions on Mobile Computing*, vol. 24, no. 7, pp. 6583–6594, 2025.
- [52] C. Fragner, A. P. Weiss, F. P. Wenzl, and E. Leitgeb, "Integrated Sensing and Communication in the Visible Spectral Range: A Novel Closed Loop Controller," in *2022 International Conference on Broadband Communications for Next Generation Networks and Multimedia Applications (CoBCom)*, 2022, pp. 1–7.
- [53] R. Zhang, Y. Shao, M. Li, L. Lu, and Y. C. Eldar, "Optical Integrated Sensing and Communication With Light-Emitting Diode," *IEEE Internet of Things Journal*, 2025.
- [54] H. Wang, Z. Zeng, C. Chen, B. Zhu, S. Shao, and M. Liu, "Retroreflective Optical ISAC Supporting 3D Positioning in Indoor Environments," in *2024 Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC)*, 2024.
- [55] P. Dong, P. Tang, Y. Yin, L. Tian, and J. Zhang, "Analysis of Visible Light-Based ISAC Channel Characteristics in a V2V Scenario," in *IEEE Wireless Communications and Networking Conference, WCNC*, 2025.
- [56] B. Zhou, A. Liu, and H. So, "Visible Light Communication-Enabled Simultaneous Position and Orientation Detection for Harnessing Multipath Interference and Random Fading," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [57] W. Wang, H. Wang, Z. Zeng, C. He, and C. Chen, "UAV-Aided Flying Retroreflective Optical ISAC with Angle Diversity Transmitters," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, 2025.
- [58] Y. Jiang, M. Xie, Z. Shi, and Y. Wang, "Monolithic III-Nitride Photonic Circuit for Integrated Sensing and Communication System," in *2023 20th China International Forum on Solid State Lighting & 2023 9th International Forum on Wide Bandgap Semiconductors (SSLCHINA: IFWS)*, 2023.
- [59] B. Jia *et al.*, "Monolithically Integrated Sensing, Communication, and Energy Harvester," *Energy Technology*, 2022.
- [60] Y. Xia *et al.*, "High-Speed Flexible Near-Infrared Organic Photodetectors for Self-Powered Optical Integrated Sensing and Communications," *Advanced Functional Materials*, vol. 35, no. 2, p. 2412813, 2025.
- [61] J. Li *et al.*, "An integrated sensing and communication solar skin for health monitoring and human-machine interaction," *Nano Energy*, vol. 148, p. 111639, 2026.
- [62] —, "Indoor Health Monitoring with VLC-based Passive Posture Monitoring," in *2024 Photonics & Electromagnetics Research Symposium (PIERS)*, 2024.
- [63] Y. Cui *et al.*, "Retroreflective optical ISAC using OFDM: channel modeling and performance analysis," *Optics Letters*, vol. 49, no. 15, p. 4214, 2024.
- [64] L. Shi, Z. Liu, B. Béchadegue, H. Guan, L. Chassagne, and X. Zhang, "Experimental Demonstration of Integrated Optical Wireless Sensing and Communication," *Journal of Lightwave Technology*, vol. 42, no. 20, pp. 7070–7084, 2024.
- [65] A. Aboltins, V. Bepal'ko, and V. Kurtenoks, "Continuous Event Recording Technology For Integrated Sensing And Communications In Space," in *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, 2024, pp. 2971–2974.
- [66] Z. Ye, L. Liu, Z. Lin, L. Liu, Z. Liu, and H. Fu, "Single-Source optical integrated sensing and communication (O-ISAC) with sequential target recognition," *Optics & Laser Technology*, vol. 198, p. 114961, 2026.
- [67] C. Carver *et al.*, "Catch Me If You Can: Laser Tethering with Highly Mobile Targets," in *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation, NSDI 2024*, 2024.
- [68] Y. Wen, F. Yang, J. Song, and Z. Han, "Joint Design of Optical Phased Array Beamforming and OFDM Waveform in Optical Wireless Integrated Sensing and Communication," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 9703–9718, 2026.
- [69] X. Shi, M. Cao, C. Chen, S. Zhang, P. Fan, and H. Wang, "OCDM Integrated LiDAR and Communication System for UAVs," in *2025 13th International Conference on Information and Communication Networks, ICIN 2025*, 2025.
- [70] T. Zhang, Z. Yan, A. Pang, and A. Dang, "Phase-Coded FMCW for Optical Integrated Sensing and Communication," in *2024 Asia Communications and Photonics Conference (ACP) and International*

- Conference on Information Photonics and Optical Communications (IPOC)*, 2024.
- [71] F. Tian, J. Zhao, Y. Zhang, G. Zheng, and T. Xu, "Gaussian Pulse-Based PSS-PPM Scheme for Optical Integrated Sensing and Communication Systems," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025.
 - [72] Y. Chen *et al.*, "Wide-Steering Integrated Sensing and Communication OPA-Based Lidar Design with Joint Waveform and Thinned Antenna," in *Opto-Electronics and Communications Conference, OECC*, 2025.
 - [73] M. Wu, H. Yang, W. Liu, Q. Yao, J. Zhang, and M. Cheriet, "An Adaptive Control System for Satellite-Ground Laser Communication Based on Integrated Sensing and Communication," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, 2025, pp. 1–4.
 - [74] N. Tihomorskis *et al.*, "Software-Defined Radio Experimental Validation of an OTFS-Based ISAC for Velocity Estimation in an ARoF Setup," *Technologies*, vol. 14, no. 5, p. 262, 2026.
 - [75] J. Wang, J. Zhang, A. P. T. Lau, and C. Lu, "Integrated ultra-large dynamic vibration sensing with fronthaul analog radio-over-fiber transmission," *Photonics Research*, vol. 13, no. 8, p. 2339, 2025.
 - [76] R. Arunachalam, R. Singh, and M. V. Kumar, "Towards last-mile connectivity in 6G-IoT: An integrated MIMO-FSO communication system and FBG sensors under atmospheric attenuations, strain and temperature effects," *Engineering Science and Technology, an International Journal*, vol. 63, p. 101958, 2025.
 - [77] A. Safriyru *et al.*, "Heterogeneous Integrated III-V-on-Soi Transmitter for 6G FiWi mmWave/FSO Integrated Sensing and Communication," in *2025 European Conference on Optical Communications (ECOC)*, 2025, pp. 1–5.
 - [78] H. Prabhakar, N. R. R. and V. A. Bohara, "A THz with VLC Assisted ISAC System Under the Impact of Human Blockages," in *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, 2026, pp. 1–6.
 - [79] X. Yu, Z. He, S. Tong, J. Meier, and T. Schneider, "Communication and Sensing via Hermite-Gaussian Time-Domain Mode-Division-Multiple-Access," in *Asia Communications and Photonics Conference, ACP*, 2025.
 - [80] Y. Z. Z. L. Wang, "Photonic generation of frequency agile LFM signals for ISAC systems," *Optical and Quantum Electronics*, 2024.
 - [81] X. Song *et al.*, "Full dimensional control of structured microwaves based on microcombs," *Nature Photonics*, vol. 20, no. 4, pp. 377–385, 2026.
 - [82] J. Shao, T. Z. Gutema, and W. O. Popoola, "Enhanced Room Coverage With Photon-Level NLOS Ultraviolet Optical Wireless Communication," *IEEE Photonics Technology Letters*, vol. 37, no. 24, pp. 1447–1450, 2025.
 - [83] I. Krikidis, "Quantum Optical Integrated Sensing and Communication With Homodyne BPSK Detection," *IEEE Wireless Communications Letters*, vol. 15, pp. 1375–1379, 2026.
 - [84] X. Ou *et al.*, "Integrated Optical Sensing and Communication (IOSAC) System Based on Hybrid-Waveguide Structures," *Advanced Materials Technologies*, vol. 9, no. 3, p. 2300998, 2024.
 - [85] M. F. Ahmed, R. U. Mondal, M. H. Kabir, and A. M. A. Rahman, "OFDM-based optical integrated sensing and communication framework using embedded pilot subcarriers," *Optics Continuum*, vol. 5, no. 6, p. 1706, 2026.
 - [86] M. Che, B. Li, H. Tang, S. Ye, Y. Kamiura, and K. Kato, "Photonic Generation of Joint Amplitude-Frequency Modulated Waveform for THz Integrated Sensing and Communication," *Microwave and Optical Technology Letters*, vol. 67, no. 5, p. e70216, 2025.
 - [87] M. Nafees, M. Baniyadi, J. R. Hopgood, M. Safari, and J. S. Thompson, "Integrated Sensing and Communication for UAV Trajectory Optimization in Mixed FSO-RF Networks in Dynamic Weather Conditions," in *2025 IEEE Wireless Communications and Networking Conference (WCNC)*, 2025, pp. 1–6.
 - [88] —, "Leveraging ISAC for Adaptive Cell Zooming in Mixed FSO-RF Networks," in *2025 IEEE 36th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2025, pp. 1–6.
 - [89] M. Ma *et al.*, "Joint break-point localization sensing based on OFDR and non-disruptive communication fiber ISAC system," *Optics & Laser Technology*, vol. 192, p. 113726, 2025.
 - [90] Y. Wen, F. Yang, J. Song, and Z. Han, "Optical Wireless Integrated Sensing and Communication Based on Optical Phased Array: Performance Metric and Optimal Beamforming," *IEEE Transactions on Wireless Communications*, vol. 24, no. 9, pp. 7221–7236, 2025.
 - [91] Z. Liu *et al.*, "CORE-Lens: Simultaneous Communication and Object REcognition with Disentangled-GAN Cameras," in *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, 2022, pp. 172–185.
 - [92] J. Shi, Y. Han, Y. Li, X. Gui, X. Fu, and Z. Li, "Common bias-digital subcarrier multiplexing scheme for integrated sensing and communication in short-reach optical transmission systems," *Optics Express*, 2025.
 - [93] Z. Chen *et al.*, "Learned Digital Back-Propagation for an optical-domain waveform integrated with communication and sensing," in *ACM International Conference Proceeding Series*, 2023.
 - [94] Y. Wen, F. Yang, J. Song, and Z. Han, "Adaptive Resource Allocation in ADO-OFDM for Optical Wireless Integrated Sensing and Communication," in *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, 2024, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
 - [95] —, "Optical Wireless Integrated Sensing and Communication Based on EADO-OFDM: A Flexible Resource Allocation Perspective," *IEEE Transactions on Wireless Communications*, 2025.
 - [96] Z. Lyu *et al.*, "Photonic Chirp Spread Spectrum THz-ISAC: Constraints and Design Considerations," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
 - [97] —, "Photonic THz Sensing-assisted Multipath Channel Estimation for ISAC," in *Optical Fiber Communication Conference (OFC 2025)*, 2025.
 - [98] B. Yang *et al.*, "Integrated Communication and Enhanced Forward Phase-Based Sensing Based on Frequency-Domain Pilot Tones in DSCM Systems Using 100 kHz ECLs," *Journal of Lightwave Technology*, 2025.
 - [99] Y. Song, M. Deng, Y. Liu, W. Wang, and F. Li, "Simultaneous polarization demultiplexing and vibration sensing via single pilot tone in DSCM-ISAC systems," *Optics Express*, 2025.
 - [100] Z. Lyu *et al.*, "Preamble-Free Synchronization Based on Dual-chirp Waveforms for Photonic THz-ISAC," *Journal of Lightwave Technology*, 2024.
 - [101] Y. Zou *et al.*, "Picosecond-Level Round-Trip Delay Measurement Scheme Based on Non-Commensurate Sampling for Future Deterministic Passive Optical Networks," *Journal of Lightwave Technology*, 2026.
 - [102] R. Kaide, S. Ye, Y. Wang, Y. Mikami, Y. Ueda, and K. Kato, "Integrated Sensing and Communication in the Terahertz Band Enabled by Photomixing Using a High-Speed Wavelength Tunable Laser," in *2025 International Symposium on Antennas and Propagation, ISAP 2025*, 2025, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
 - [103] K. Matsuda *et al.*, "Sensing μm -scale vibrations in the Hz range within a THz communication system," *Optics Express*, 2026.
 - [104] B. Yin *et al.*, "Integrated Sensing and Communication in Self-Homodyne MCF Transmission Systems With Linear Frequency Modulation," *Journal of Lightwave Technology*, 2026.
 - [105] M. Salhi and M. Sliti, "Secure Multi-Tenant Distributed Sensing for ISAC-Over-Fiber (IOSAC) Networks," *IEEE Access*, 2026.
 - [106] M. Lei *et al.*, "Photonics-assisted THz ISAC System Based on A Time-Frequency Efficient SF-LFM-OFDM Waveform," in *2026 Optical Fiber Communications Conference and Exhibition (OFC)*, 2026, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
 - [107] Y. Tao *et al.*, "Integrated photonic ultrawideband real-time spectrum sensing for 6G wireless networks," *Nature Communications*, 2026.
 - [108] S. Chen, Z. Zeng, and C. Chen, "Experimental Demonstration of Multi-User Full-Duplex RO-ISAC System," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025.
 - [109] Z. Lyu *et al.*, "A Beam-Scannable Photonic THz-ISAC System Based on Risley Prisms," in *2023 Asia Communications and Photonics Conference/2023 International Photonics and Optoelectronics Meetings (ACP/POEM)*, 2023, pp. 1–4.
 - [110] M. Wang, W. Zhang, Z. Ren, S. Li, X. Zheng, and X. Xue, "Frequency-comb-steered ultrawideband quasi-true-time-delay beamformer for integrated sensing and communication," *Nature Communications*, vol. 16, no. 1, p. 7571, 2025.
 - [111] Y. Men, J. Ji, A. Wen, Y. Wang, and F. Yang, "Photonic-assisted generation of joint radar and communication signals with immunity to power fading," *Optics Communications*, 2023.
 - [112] L. Peng, M. Yin, D. Zou, N. Yang, Y. Xiao, and F. Li, "Photonics-assisted integrated sensing and communication with ranging resolution improvement by multiple signal classification," *Optics Express*, 2024.

- [113] W. Bai *et al.*, "Millimeter-wave joint radar and communication system based on photonic frequency-multiplying constant envelope LFM-OFDM," *Optics Express*, 2022.
- [114] Y. Wen, F. Yang, J. Song, and Z. Han, "Free Space Optical Integrated Sensing and Communication Based on LFM and CPM," *IEEE Communications Letters*, 2024.
- [115] S. Chen *et al.*, "Microwave Photonics Empowered Minimalist Base Station for Integrated Sensing and Communication via Cascaded Intensity and Phase Modulation," *Journal of Lightwave Technology*, 2026.
- [116] N. Zhong, P. Li, W. Bai, W. Pan, L. Yan, and X. Zou, "Spectral-Efficient Frequency-Division Photonic Millimeter-Wave Integrated Sensing and Communication System Using Improved Sparse LFM Sub-Bands Fusion," *Journal of Lightwave Technology*, 2023.
- [117] L. Peng, D. Luo, Y. Xiao, and F. Li, "A Photonics-aided MMW OFDM Joint Radar and Communication System with Velocity Accuracy Improvement," in *APSCON 2023 - IEEE Applied Sensing Conference, Symposium Proceedings*, 2023, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [118] B. Dong *et al.*, "Demonstration of photonics-based flexible integration of sensing and communication with adaptive waveforms for a W-band fiber-wireless integrated network," *Optics Express*, 2022, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [119] J. Gao, D. Liang, T. Shi, and Y. Chen, "Microwave Photonic Integrated Sensing and Communication Based on Polarization Multiplexing and Frequency-to-Time Mapping," *IEEE Internet of Things Journal*, 2025.
- [120] Z. Wang *et al.*, "Co-Route Fiber Recognition and Status Diagnosis Based on Integrated Sensing and Communication in 6G Transport Networks," *IEEE Internet of Things Journal*, vol. 11, no. 18, pp. 29 348–29 359, 2024.
- [121] S. Liu, T. Sun, L. Wang, J. Lin, J. Song, and Y. Dong, "Integrated PC-FMCW Automotive Headlamp for Communication, Sensing, and Illumination," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025.
- [122] Z. Xie, W. Dong, Z. Zhu, F. Deng, K. Yin, and T. Jiang, "A Novel Comb Based Multi-Function Architecture for ISAC System," *Journal of Lightwave Technology*, 2026.
- [123] Z. Wei *et al.*, "Simultaneous distributed sensing and communication enabled by space and wavelength division multiplexing in a field-deployed seven-core fiber," *Journal of Optical Communications and Networking*, vol. 18, no. 4, p. B49, 2026.
- [124] Y. Hu *et al.*, "Demonstration of High Sensitivity Detection and High Precision Localization of Vibrations in Bidirectional Coherent Transmission Systems Enabled by Carrier Laser Phase Noise Retrieval," *Journal of Lightwave Technology*, 2025.
- [125] Y. Xu, T. Wang, P. Huang, and G. Zeng, "Integrated Distributed Sensing and Quantum Communication Networks," *Research*, 2024.
- [126] X. Li *et al.*, "Integrated Sensing and Communication Chip Based on III-Nitride for Motion Detection," *ACS Omega*, vol. 8, no. 16, pp. 14 656–14 661, 2023.
- [127] Y. Zhu *et al.*, "Long-Distance Integrated Forward Vibration Sensing and Terabit Optical Communication with 12.5-MHz DFB Laser via Phase Noise Separation," in *2026 Optical Fiber Communications Conference and Exhibition (OFC)*, 2026.
- [128] X. Meng *et al.*, "Photonics-Assisted System for Joint Radar Ranging, Wireless Communications, and Spectrum Sensing," *Journal of Lightwave Technology*, 2026.
- [129] T. Wei, S. Liu, and X. Du, "Visible Light Integrated Positioning and Communication: A Multi-Task Federated Learning Framework," *IEEE Transactions on Mobile Computing*, vol. 22, no. 12, pp. 7086–7103, 2023.
- [130] X.-X. Feng, M.-Y. Han, M.-P. Chen, Q. Fang, Y.-J. Wang, and X. Li, "Integrated Nitride optoelectronic chip for motion detection and visible light communication," *Chinese Optics*, 2023, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [131] B. Yang, Q. Zhuo, H. Yu, J. Tang, S. Gao, and Y. Yang, "Vibration Localization under Unidirectional Single-Channel Transmission Conditions in the DSCM Systems Based on Non-Reciprocity," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025.
- [132] M. Fang *et al.*, "On-Chip Photonic MMW Joint Communication and Radar System," in *2023 International Topical Meeting on Microwave Photonics (MWP)*, 2023.
- [133] J. Chen *et al.*, "Photonic-Assisted Joint Radar and Communication System Using Overlapped Spectrum for High-Resolution Ranging and High-Speed Data Transmission," *IEEE Transactions on Microwave Theory and Techniques*, 2025.
- [134] R. T. R. W. G. J. X. H. L. Z. H. Z. Wu, "Seamless integration of distributed acoustic sensing and passive optical networks for human intrusion monitoring," *Communications Engineering*, 2026.
- [135] T. Chu *et al.*, "Revolutionizing 6G: Experimental Validation of an Optical Integrated Communication, Sensing, and Power Transfer System," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 689–701, 2026.
- [136] L. Wang *et al.*, "Endogenous integration of communication and interference fading free sensing using telecom pilots via joint polarization-fraction domain multiplexing," *Optics Express*, 2025.
- [137] Y. Wang, L. Wang, Z. Lin, C. Lu, M. Tang, and C. Yu, "Endogenous ISAC Enabled Interference Fading Free Sensing via Telecom Training Sequence," in *Opto-Electronics and Communications Conference, OECC*, 2025, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [138] Y. Wang, L. Wang, Y. Li, C. Yu, M. Tang, and Z. Wei, "Integration of In-band Polarization-Based Sensing and Communication via Fraction Domain Multiplexing," in *Asia Communications and Photonics Conference, ACP*, 2025, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [139] F. Liu *et al.*, "Millimeter-wave over fiber integrated sensing and communication system using self-coherent OFDM," *Optics Express*, 2024.
- [140] F. Liu, X. Zou, N. Zhong, X. Deng, L. Yan, and W. Pan, "Signal-Signal Beating Interference: From Destructive to Constructive for Photonic THz Integrated Sensing and Communication System Using Self-Coherent OFDM," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [141] B. Dong *et al.*, "Photonic-Based Flexible Integrated Sensing and Communication With Multiple Targets Detection Capability for W-Band Fiber-Wireless Network," *IEEE Transactions on Microwave Theory and Techniques*, 2024.
- [142] Y. Zuo, W. Liu, Y. Cheng, X. Li, and M. Luo, "High-performance Vibration Sensing Based on Subband Phase-Shift Transform in Integrated Sensing and Communication System," in *Asia Communications and Photonics Conference, ACP*, 2025.
- [143] J. Wang, A. P. T. Lau, and C. Lu, "Sensing as a Service: Can In-Band Distributed Fiber Optic Sensing Coexist With Coherent Metro DWDM Communication Network?" *Journal of Lightwave Technology*, 2026.
- [144] B. Dong *et al.*, "THz Integrated Sensing and Communication With Full-Photonic Direct LFM Reception and De-Chirping for D-Band Fiber-Wireless Network," *IEEE Transactions on Microwave Theory and Techniques*, 2025.
- [145] Q. Zhou *et al.*, "A Sub-THz ISAC System with Simultaneous Real-Time 125.516-Gbps Communication Rate and Offline 10-mm Sensing Resolution Enabled by Photonics," in *ECOC 2024: 50th European Conference on Optical Communication*, 2024.
- [146] W. Bai *et al.*, "Photonic Millimeter-Wave Joint Radar Communication System Using Spectrum-Spreading Phase-Coding," *IEEE Transactions on Microwave Theory and Techniques*, 2022.
- [147] X. Liu *et al.*, "Integrating High-Capacity Self-Homodyne Transmission and High-Sensitivity Dual-Pulse ϕ -OTDR with an EO Comb over a 7-Core Fiber," *Photonics*, vol. 13, no. 3, p. 261, 2026.
- [148] B. Yang *et al.*, "Accurate vibration recognition enabled by synergistic sensing based on both phase and SOP for stable IM-DD optical interconnects," *Journal of Optical Communications and Networking*, 2026.
- [149] M. Lei *et al.*, "Integration of Sensing and Communication in a W-Band Fiber-Wireless Link Enabled by Electromagnetic Polarization Multiplexing," *Journal of Lightwave Technology*, 2023.
- [150] Y. Wen, F. Yang, J. Song, and Z. Han, "Interference Cancellation for Optical Wireless ISAC Based on Optical Phased Array," in *GLOBE-COM 2025 - 2025 IEEE Global Communications Conference*, 2025.
- [151] X. Xie *et al.*, "Adaptive Lighting Control in Visible Light Systems: An Integrated Sensing, Communication, and Illumination Framework," in *2025 11th International Conference on Computer and Communications (ICCC)*, 2025.
- [152] T. Shi, F. Zhang, and Y. Chen, "Microwave Photonic Joint Radar and Secure Communication via Radar Signal Masking," *IEEE Microwave and Wireless Technology Letters*, 2025.
- [153] S. Chen, C. Chen, Z. Zeng, Y. Yang, and H. Haas, "Full-Duplex RO-ISAC System: Wavelength Division Duplexing and Hybrid Waveform Design," *IEEE Photonics Technology Letters*, 2025.
- [154] T. Chu *et al.*, "Flexible Hybrid Waveform Design for Optical Integrated Communication, Sensing, and Power Transfer Systems," *Journal of Lightwave Technology*, 2026.

- [155] B. Yang, J. Sun, H. Chi, K. Xu, and S. Yang, "Joint radar and communication system based on a chaotic optoelectronic oscillator," *Optics Communications*, 2024.
- [156] M. A. Allousch and A. Sandmann, "Cc-Otdr Sequence Shaping Enabling Joint Co-Directional Sensing and Communication," in *2025 IEEE Photonics Conference (IPC)*, 2025, pp. 1–2.
- [157] L. Tang, Z. Hu, D. Teng, and L. Liu, "Integrated high precision time-of-flight depth sensing and optical communication based on Hamiltonian coding," *Optics Communications*, 2023.
- [158] T. Wang *et al.*, "Integrated secure communication and sensing based on simultaneous chaotic encryption and vibration detection with an RE-LFM signal," *Journal of Optical Communications and Networking*, 2026.
- [159] M. Lei *et al.*, "Demonstration of Photonic Sub-THz ISAC System with Real-Time 251.03-Gbps Communication Rate and Offline 2.5-cm Sensing Resolution," in *2024 22nd International Conference on Optical Communications and Networks (ICOON)*, 2024.
- [160] Z. Wang *et al.*, "Probabilistic Shaping With Bit-Weighted Distribution Matching for Photonics-Assisted mmWave Dual-Functional OFDM ISAC Systems," *Journal of Lightwave Technology*, 2026.
- [161] J. Y. Wang, H. N. Yang, J. B. Wang, M. Lin, and P. Shi, "Joint Optimization of Slot Selection and Power Allocation in Integrated Visible Light Communication and Sensing Systems," *IEEE Internet of Things Journal*, 2023.
- [162] M. Liu, J. Wang, C. Liu, C. Yu, and C. Lu, "Coexistence of Distributed Sensing and Ultra-Simple Coherent RoF Based Front-Haul for Radio Access Network," *Journal of Lightwave Technology*, 2025.
- [163] M. Lei *et al.*, "Photonics-aided integrated sensing and communications in mmW bands based on a DC-offset QPSK-encoded LFM CW," *Optics Express*, vol. 30, no. 24, p. 43088, 2022.
- [164] F. Liu, G. Zhu, and X. Zhou, "Enhanced single-fiber bi-directional FT-DVS using a 600-kHz linewidth DBR laser with LPN compensation for O-ISAC [Invited]," *Chinese Optics Letters*, 2026.
- [165] G. Chen *et al.*, "Deep Integration Between Polarimetric Forward-Transmission Fiber-Optic Communication and Distributed Sensing Systems," *Sensors*, 2024.
- [166] R. Zhu *et al.*, "Deep Integration of Fiber-Optic Communication and Sensing Systems Using Forward-Transmission Distributed Vibration Sensing and on-off Keying," *Sensors*, 2024.
- [167] J. C. Wang, T. Y. Liu, and J. H. Zhou, "Fiber Events Recognition in Integrated Sensing and Communication Systems Based on Signal State of Polarization Analysis," in *2025 Photonics & Electromagnetics Research Symposium - Fall (PIERS-Fall)*, 2025, pp. 1–6.
- [168] M. Lei *et al.*, "Photonics-Aided THz ISAC System Based on A Spectral Overlapping Multi-Carrier 16QAM-LFM Waveform," in *2025 Optical Fiber Communications Conference and Exhibition, OFC 2025 - Proceedings*, 2025.
- [169] X. Yu *et al.*, "Photonics-assisted self-interference cancellation for in-band full-duplex integrated sensing and communication transceiver," *Optics Express*, 2024.
- [170] Z. Yixiao, Z. Shanghong, L. Xuan, W. Guodong, and Y. Longqiang, "Photonic-aid dual-formats LFM signals generator for joint radar-communication system," *Optik*, 2022.
- [171] H. Wang, C. Chen, Z. Zeng, S. Shao, and H. Haas, "Bidirectional Retroreflective Optical ISAC Using Time Division Duplexing and Clipped OFDM," *IEEE Photonics Technology Letters*, 2025.
- [172] W. Wang *et al.*, "Multiuser Retroreflective Optical ISAC: Waveform Design, Joint Sensing and Multiple Access," *Journal of Lightwave Technology*, 2026.
- [173] G. Zhou, M. Cao, Q. Yang, Y. Zhang, X. Zhang, and H. Wang, "Bi-LSTM-Transformer hybrid network for V2V optical wireless ISAC systems," *ICT Express*, 2026.
- [174] B. Yang, J. Tang, S. Gao, J. Liang, J. Wei, and Y. Yang, "Flexible Vibration Sensing Using Kalman Smoother Integrated in ECL-Based Communication Systems," *IEEE Photonics Technology Letters*, 2025.
- [175] Y. Liu, X. Deng, N. Zhong, X. Zou, W. Pan, and L. Yan, "Simulation of Constant-Envelope THz Integrated Sensing and Communication System Based on Photonics with 2D Positioning," in *2024 Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC)*, 2024.
- [176] S. Migla *et al.*, "Demonstration of Integrated Sensing and Communications in an AROF-Based Radar System Utilizing Pulse Position Modulation," *IEEE Access*, 2026.
- [177] J. Zhang *et al.*, "Photonics-Aided THz Integrated Sensing and Communication System Based on a Time-Frequency Efficient SF-LFM-OFDM Waveform," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [178] Q. Wang, Y. Deng, D. Mishra, Y. Xie, E. Aboutanios, and S. Atakaramians, "Photonics-based Terahertz Integrated Communication and Localization for D-band," in *2024 49th International Conference on Infrared, Millimeter, and Terahertz Waves (IRMMW-THz)*, 2024.
- [179] L. Li *et al.*, "THz-Over-Fiber System With Orthogonal Chirp Division Multiplexing for Integrated Sensing and Communication," *Journal of Lightwave Technology*, 2024.
- [180] B. Yang, J. Tang, L. Fan, J. Wei, and Y. Yang, "Machine-learning-enhanced phase-based vibration sensing in coherent DSCM systems under commercial ECLs," *Science China Information Sciences*, vol. 69, no. 6, p. 160306, 2026.
- [181] Z. Xue, S. Li, X. Xue, X. Zheng, and B. Zhou, "Photonics-assisted joint radar and communication system based on an optoelectronic oscillator," *Optics Express*, vol. 29, no. 14, p. 22442, 2021.
- [182] Y. Hai, Y. Luo, C. Liu, and A. Dang, "Remote Phase-Shift LiDAR With Communication," *IEEE Transactions on Communications*, vol. 71, no. 2, pp. 1059–1070, 2023.
- [183] W. Liu, Q. Deng, Z. Yang, F. Xiao, B. Zhou, and X. Geng, "Integrated Optical Camera Communication and Scene Sensing Based on Generative Adversarial Networks," *IEEE Transactions on Mobile Computing*, vol. 25, no. 6, pp. 9010–9025, 2026.
- [184] Z. Yixiao *et al.*, "Photonic-aided Doppler-robust ISAC system based on index modulation and chirp multiplexing," *Optics Express*, vol. 33, no. 17, p. 35059, 2025.
- [185] D. Du, Y. Wang, and Y. Gao, "Integrated Communication and Radar Detection Based-on Single Modulator at W-Band," in *2024 7th International Conference on Information Communication and Signal Processing (ICICSP)*, 2024, pp. 1158–1162.
- [186] A. Bansal, A. Bhatnagar, V. Sharma, and N. Sharma, "Optical integrated sensing and communication (O-ISAC): A unified framework for 6G and AI-Powered intelligent networks," *Journal of Optical Communications*, 2026.
- [187] K. Deng, J. Liu, X. Lu, J. Chen, Y. Zhou, and W. Li, "A Unified OFDM-ISAC Signal Generation Architecture in W-Band via Photonics-Aided Frequency Multiplication and Phase Noise Mitigation," *Photonics*, vol. 12, no. 11, p. 1052, 2025.
- [188] J. Liu *et al.*, "W-band photonics-aided OFDM system integrating sensing and communication with phase noise suppression scheme," *Optics and Laser Technology*, 2025.
- [189] H. Dong, K. Xi, H. Song, S. Liu, and Z. Zhang, "An Information Interaction Method for Autonomous Train Operation Protection Based on Detection and Communication Integrated Microwave Photonic Radar," *IEEE Transactions on Instrumentation and Measurement*, 2026.
- [190] L. Shi, B. Bechadargue, L. Chassagne, and H. Guan, "Joint Visible Light Sensing and Communication Using m-CAP Modulation," *IEEE Transactions on Broadcasting*, 2023.
- [191] H. Wang, Z. Zhang, Y. Ge, and B. Zhu, "Optical Integrated Sensing and Communication System Based on Combination of OIRS and PD Array for Mobile Scenarios," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 19 522–19 536, 2024.
- [192] H. A. Satai, B. Bechadargue, L. Chassagne, K. Acuna-Condori, B. Cagneau, and P. Ruau, "Single-Source Integrated Optical Wireless Positioning and Communication," in *2026 IEEE 6th International Symposium on Joint Communications & Sensing (JC&S)*, 2026, pp. 1–6.
- [193] X. Xiong, H. Wang, S. Shao, and C. Chen, "Hybrid MRR-Assisted Long-Range SDMA for RO-ISAC Systems in Industry IoT Applications," *Journal of Lightwave Technology*, 2026.
- [194] Z. Yue *et al.*, "Demonstration of Ranging-Based Phase Encryption for Enhanced Free-Space Optical Security," *Journal of Lightwave Technology*, 2026.
- [195] F. Yang, C. Lin, J. Gao, and Y. Chen, "Microwave Photonic Joint Radar, Wireless Communications, and Spectrum Sensing System With Broadband Tunability From 12 to 40 GHz," *IEEE Internet of Things Journal*, 2026.
- [196] Y. Wen, F. Yang, J. Song, and Z. Han, "Pulse Sequence Sensing and Pulse Position Modulation for Optical Integrated Sensing and Communication," *IEEE Communications Letters*, 2023.
- [197] Q. Bai and J. Ma, "Photonic-assisted generation of fast switchable multi-format LFM waveforms with a tuned bandwidth for a joint radar-communication system," *Optics Communications*, 2025.
- [198] Z. Wang *et al.*, "Simultaneous Full-Photonic Dechirping and Demodulation THz-ISAC System Based on Thin-Film Lithium Niobate Modulator," *Journal of Lightwave Technology*, vol. 44, no. 11, pp. 4640–4647, 2026.

- [199] P. Munar-Vallespir, J. Nötzel, and F. Seitz, "Joint Communication and Eavesdropper Detection on the Lossy Bosonic Channel," in *Proceedings - IEEE Global Communications Conference, GLOBECOM*, 2024.
- [200] B. Li, J. Wang, Y. Gao, J. Lian, and B. Lian, "LACO-OFDM-Based Optical ISAC System Design and Performance Analysis," *IEEE Internet of Things Journal*, 2026.
- [201] Z. Wang *et al.*, "Bit Inversion Probabilistic Shaping Enabling Performance Tradeoff in Photonics-Aided Millimeter-Wave OFDM Integrated Sensing and Communication Systems," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [202] Z. Lyu *et al.*, "Radar-Centric Photonic Terahertz Integrated Sensing and Communication System Based on LFM-PSK Waveform," *IEEE Transactions on Microwave Theory and Techniques*, 2023.
- [203] J. Shen, J. Fan, M. Li, Y. Fei, T. Fu, and M. Gao, "Multi-service integrated sensing and communication system for industrial internet-of-things," *Optics and Lasers in Engineering*, 2026.
- [204] J. He, K. Liu, L. Yin, and Z. Chen, "An Orthogonal Time Frequency Space Waveform With W-MUSIC Scheme for Photonic-Aided ISAC System," *IEEE Photonics Technology Letters*, 2026.
- [205] J. He, L. Yin, K. Liu, S. Xiao, and Z. Chen, "Zero-Padding 2D-DFT Scheme for Photonics-Assisted W-Band Fiber-Wireless ISAC System With Orthogonal Time Frequency Space Modulation," *Journal of Lightwave Technology*, 2025.
- [206] Y. Wu *et al.*, "Advanced Integration of Sensing and Communication With High DSP Compatibility for SMART Network," *IEEE Photonics Technology Letters*, vol. 37, no. 21, pp. 1249–1252, 2025.
- [207] H. Xiao, W. Wang, K. Zhang, F. Li, and C. Sun, "Microalgae Density Measurement Using Quantum-Dot-Integrated Sensing and Communication System," *IEEE Transactions on Instrumentation and Measurement*, 2025.
- [208] M. Anastasopoulos, J. Gutiérrez, and A. Tzanakaki, "Optical Transport Network Optimization Supporting Integrated Sensing and Communication Services," in *Optical Fiber Communication Conference in Proceedings Optical Fiber Communication Conference, OFC 2025 and Optical Fiber Communication Conference (OFC) Postdeadline Papers 2025*, 2025.
- [209] N. Zhong *et al.*, "Photonics Terahertz Dual-Band Anti-Jamming Integrated Sensing and Communication System Empowered by Permutation Frequency Hopping," *Journal of Lightwave Technology*, 2026.
- [210] J. Zhang *et al.*, "Photonics-Aided THz Integrated Sensing and Communication System Based on a Subcarrier-Chirp Inter-Embedded Waveform," *IEEE Open Journal of the Communications Society*, 2025.
- [211] B. Li *et al.*, "Passive Sensing Technique Using Cyclic Prefix in Power-Constrained Optical Wireless Communication Systems," *IEEE Transactions on Vehicular Technology*, 2025.
- [212] G. Zhou, M. Cao, Q. Yang, Y. Zhang, S. Han, and H. Wang, "Integrated Design of OFDM-LFM Lidar Communication in IoV Environment," in *2024 IEEE 12th International Conference on Information, Communication and Networks (ICICN)*, 2024.
- [213] B. Zhou, X. Wang, Y. Shen, and P. Fan, "6-DoF Location-and-Pose Estimation Toward Integrated Visible Light Communication and Sensing: Algorithm Design and Performance Limits," *IEEE Transactions on Signal Processing*, 2024.
- [214] Y. Zhu *et al.*, "Integrated 1-Tb/s/λ Coherent Optical Transmission and Forward Vibration Sensing Using DFB Lasers with Combined Linewidth of 21.4 MHz," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–5.
- [215] J. Wu, Z. Liu, L. Liu, H. Ji, and W. Shieh, "Simultaneous Transmission and Sensing Emulation Using Interconnected Counter-Propagating Recirculating Loops," in *2025 European Conference on Optical Communications (ECOC)*, 2025.
- [216] A. G. Khorasgani, M. Mirmohseni, and A. Elzanaty, "Optical ISAC: Fundamental Performance Limits and Transceiver Design," in *2024 IEEE Globecom Workshops (GC Wkshps)*, 2024.
- [217] J. He, J. Li, K. Liu, S. Xiao, and Z. Chen, "A Cross-Correlation MUSIC Scheme Based on OFDM for Photonics-Aided W-Band ISAC Systems," *IEEE Photonics Technology Letters*, 2026.
- [218] Z. Saghir and S. Aboagye, "Movable Access Point-Aided Integrated Visible Light Communication and Sensing Networks," *IEEE Wireless Communications Letters*, 2025.
- [219] H.-N. Yang, J.-Y. Wang, Q. Zeng, M. Cheng, M. Lin, and J.-B. Wang, "Joint Task Offloading and Resource Allocation for Integrated VLC and Sensing in Digital Twin-Aided Vehicular Edge Computing Networks," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 6360–6372, 2025.
- [220] T. Li *et al.*, "Submarine integrated sensing and communication cable networks enable high-precision earthquake location estimation through linear wavefront theory," *Journal of Optical Communications and Networking*, vol. 18, no. 4, p. B108, 2026.
- [221] X. Zhang *et al.*, "Microcomb-synchronized optoelectronics," *Nature Electronics*, 2025.
- [222] J. Jia, B. Dong, L. Tao, J. Shi, N. Chi, and J. Zhang, "Demonstration of radar-aided flexible communication in a photonics-based W-band distributed integrated sensing and communication system for 6G," *Chinese Optics Letters*, 2024.
- [223] F. Li *et al.*, "Double-effect DNN-based DBP Scheme for Integrated Sensing and Communications (ISAC)," in *2022 Optical Fiber Communications Conference and Exhibition (OFC)*, 2022.
- [224] F. Zhang *et al.*, "Glass fiber-based integrated sensing and communication system for vehicular applications," *Advanced Photonics Nexus*, 2026.
- [225] J. Wu, X. Lin, C. Yang, S. Yang, C. Liu, and Y. Cao, "Wearable Sensors Based on Miniaturized High-Performance Hybrid Nanogenerator for Medical Health Monitoring," *Biosensors*, 2024.
- [226] Z. Jiang *et al.*, "Optoelectronic integrated sensing and communication system based on InGaN/GaN MQW for wireless detection of air pressure," *Sensors and Actuators A: Physical*, vol. 395, p. 117086, 2025.
- [227] A. Rode *et al.*, "Machine learning opportunities for integrated polarization sensing and communication in optical fibers," *Optical Fiber Technology*, vol. 90, p. 104047, 2025.
- [228] X. Cao *et al.*, "Intelligent submarine environmental monitoring based on fiber-optic integrated sensing and communication system," *Science China Information Sciences*, vol. 69, no. 6, p. 160303, 2026.
- [229] E. Ip, Y.-K. Huang, M.-F. Huang, F. Yaman, J. Fang, and T. Wang, "Integrated sensing and communications over deployed terrestrial and submarine telecom cables," *Science China Information Sciences*, vol. 69, no. 6, p. 160304, 2026.